

The Three Rationalities

**Theory, Existing Tests, and How to Build a New One That Goes Beyond
Them**

Rationality over Time · Rationality under Risk · Rationality of Machines

A working reference prepared for Alessandro Facchini

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Abstract

This is a self-contained study reference. Its purpose is to let you (i) understand each of the three theories deeply and as they stand *today*, (ii) know exactly which empirical tests already exist and what they assume, and (iii) see, concretely, how to build a *new, more robust test* that goes beyond them — in the spirit of testing a deductive theory through an instrument Z orthogonal to confounding shocks (your GARP-conditional-on- Z work; Adão–Costinot–Donaldson). Chapter 1 is the shared toolkit (revealed preference, identification, inference, welfare). Chapters 2–4 treat the three theories on a common template: *theory* $\rightarrow$ *testable predictions* $\rightarrow$ *existing tests* $\rightarrow$ *the gap* $\rightarrow$ *your new test*. Chapter 5 synthesizes. Each chapter ends with a short, ordered reading list of primary sources. Read it slowly; it is meant to be the thing you study before writing the proposal.

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Chapter 1

Foundations: the method you will reuse three times

This chapter builds the toolkit that the three theory chapters all draw on. If you internalize it, each later chapter becomes an application of the same five moves: (1) state a deductive theory; (2) turn it into *inequalities* on observable choices; (3) confront those inequalities with data *through an instrument* Z that is orthogonal to confounding shocks; (4) do correct *inequality inference*; (5) translate the verdict into *welfare and policy*.

1.1 Deductive vs. inductive: what makes a theory testable

The distinction that defines this whole program

A theory is worth *testing* when its prediction is *derived* from assumptions — a theorem, an equilibrium condition, an axiom system — so that data could *contradict* it. Re-checking a relationship that was itself *estimated* from data (an elasticity, a regression coefficient) on a new sample is out-of-sample validation, not a test of a theory: the “theory” is only a compressed description of past data. Your three targets — rational intertemporal choice, expected utility, and utility maximization by AI agents — are all *deductive*, and each is testable by revealed preference.

The practical consequence is that deductive theories almost always deliver *inequalities*, not equations. “A utility maximizer’s choices contain no preference cycle” is an inequality system; “the elasticity is -0.4 ” is a point estimate. This is why the inference tools below are about *moment inequalities*, not ordinary regression.

1.2 Revealed preference I: utility maximization and GARP

We observe a consumer (or an algorithm, or a household) making T choices. In observation t she faces prices $\mathbf{p}^t \in \mathbb{R}_{++}^n$ and chooses bundle $\mathbf{x}^t \in \mathbb{R}_+^n$ from the budget $\{\mathbf{x} : \mathbf{p}^t \cdot \mathbf{x} \leq \mathbf{p}^t \cdot \mathbf{x}^t\}$. The data set is $\mathcal{O} = \{(\mathbf{p}^t, \mathbf{x}^t)\}_{t=1}^T$.

Definition 1.1 (Revealed preference). $\mathbf{x}^t$ is *directly revealed preferred* to $\mathbf{x}$, written $\mathbf{x}^t \succsim^R \mathbf{x}$, if it was chosen when $\mathbf{x}$ was affordable: $\mathbf{p}^t \cdot \mathbf{x} \leq \mathbf{p}^t \cdot \mathbf{x}^t$. Let $\succsim^{R*}$ be the transitive closure of $\succsim^R$ (chains of directly-revealed preference). The strict version $\succ^R$ holds when $\mathbf{x}$ was strictly cheaper.

Definition 1.2 (GARP). $\mathcal{O}$ satisfies the *Generalized Axiom of Revealed Preference* if $\mathbf{x}^s \succsim^{R*} \mathbf{x}^t$ implies *not* $\mathbf{x}^t \succ^R \mathbf{x}^s$. In words: there are no preference cycles in which some link is strict.

Theorem 1.1 (Afriat 1967; Diewert 1973; Varian 1982). *The following are equivalent: (i) $\mathcal{O}$ is rationalized by some locally non-satiated utility; (ii) $\mathcal{O}$ satisfies GARP; (iii) there exist numbers U^t and $\lambda^t > 0$ (“Afriat numbers”) with*

$$U^s \leq U^t + \lambda^t \mathbf{p}^t \cdot (\mathbf{x}^s - \mathbf{x}^t) \quad \text{for all } s, t$$

(iv) $\mathcal{O}$ is rationalized by a continuous, increasing, concave utility.

Two lessons sit inside this theorem and recur everywhere. First, the empirical content of “maximizes a (concave) utility” is *exactly* GARP — nothing more, nothing less; the concavity is free. Second, condition (iii) is a *finite system of linear inequalities* (the **Afriat inequalities**): checking the theory is a linear-feasibility problem, and any *strengthening* of rationality (expected utility, discounting, ...) will appear as *more* linear inequalities of the same flavour. That is why one toolkit serves all three theories.

Definition 1.3 (CCEI — measuring *how badly* the theory fails). The *Critical Cost Efficiency Index* $e \in [0, 1]$ (Afriat 1972) is the largest number such that the data satisfy GARP when each budget is shrunk to $e \mathbf{p}^t \cdot \mathbf{x}^t$. $e = 1$ is exact consistency; $e = 0.95$ means the consumer “wasted” up to 5% of budget relative to a perfectly consistent plan. CCEI is the standard severity index, but note: *by itself it has no sampling theory* — a point we return to under inference and power.

1.2.1 Revealed-preference bounds (counterfactual prediction)

GARP also lets you *predict* nonparametrically. Given a price $\mathbf{p}^{T+1}$, the set of bundles the consumer could choose *without* violating GARP is the **revealed-preference bound** — a prediction interval that uses rationality alone, no functional form (Varian 1982; Blundell–Browning–Crawford 2008). This is the engine in the final Monte-Carlo pipeline (`code/final/`, Section 2.14), and it generalizes to risk and time.

1.3 Revealed preference II: the ladder of stronger rationalities

Each specific theory is a *strengthening* of GARP, and — remarkably — each implies the same qualitative signature: an *inverse relation between prices and quantities* (Echenique 2020). The ladder, from weakest to strongest:

1. **Utility maximization** (any stable preference) $\Leftrightarrow$ GARP (Afriat).
2. **Expected utility under risk** (objective probabilities): choices of state-contingent payoffs must satisfy the *Strong Axiom of Revealed Objective Expected Utility* (Kubler–Selden–Wei 2014; Polisson–Quah–Renou 2020).
3. **Subjective expected utility** (unknown probabilities): stronger still (Echenique–Saito 2015); **maxmin EU** for ambiguity (Gilboa–Schmeidler model; characterized by Chambers–Echenique–Saito 2016).
4. **Exponentially / quasi-hyperbolically discounted utility** over time (Browning 1989; Crawford 2010; Blow–Browning–Crawford 2021).

Intuition — Why “stronger theory” means “more inequalities”

GARP says the data fit *some* utility. Expected utility says they fit a utility of a *special separable form* (probability-weighted sum). Discounted utility says they fit a *special intertemporal* form. Each restriction carves the data more finely, so each adds inequalities. A test of the stronger theory is therefore a test of “GARP plus extra structure,” and a rejection can be *localized*: did the within-period part fail (basic rationality) or the extra structure (the EU/discounting part)? Crawford (2010) and Blow–Browning–Crawford

(2021) exploit exactly this separation.

1.4 Identification I: the confounding-shock problem

Here is the obstacle that motivates your whole approach. Observed behaviour is driven by two forces at once.

The two forces (Adão–Costinot–Donaldson 2025)

Write an observed change as

$$\underbrace{\Delta y}_{\text{data}} = \underbrace{\Delta x}_{\text{force the theory is about}} + \underbrace{\Delta \eta}_{\text{all other shocks}}$$

A naïve test compares the theory’s prediction to Δy and so *cannot separate* Δx from $\Delta \eta$. It can reject a correct theory (because $\Delta \eta$ is large) or accept a wrong one (because $\Delta \eta$ happens to align). Standard “goodness-of-fit” (R^2 , correlation of predicted vs. observed) thus *mixes testing with forecasting*: a model may fit well for reasons that have nothing to do with whether its causal claim is right.

In each of the three theories the nuisance has a concrete identity: *serially-correlated taste shocks* (time), *selection into who gambles/insures* (risk), and *prompt/context correlated with the task* (machines). Each masquerades as evidence for or against the theory.

1.5 Identification II: the instrument Z (your core move)

Project onto a Z orthogonal to the nuisance

Find variation Z that is (a) *correlated with the force of interest* (the price, the budget, the availability of risk) but (b) *orthogonal to the nuisance shocks*. Test the theory only on the part of the data that moves with Z . If the theory is correct, its prediction error must be *uncorrelated with Z* :

$$\mathbb{E}[Z \cdot (\text{prediction error})] = 0.$$

ACD build Z as a *shift-share* of exogenous policy shifters and test this moment; you build Z as exogenous price/budget variation unrelated to tastes and test rationality *on the Z -driven variation*. Independent noise no longer triggers rejection; only systematic, Z -correlated violations do.

Example 1.1 (The template, stated once). Test GARP, but worry that tastes shift with prices (people “want what is on sale”). If taste shifts are independent of prices, GARP violations are noise. If they are price-correlated, a naïve GARP test is fooled. *Fix*: use price variation driven by Z (a randomized menu, an announced tax, a legalization date) that is exogenous to tastes, and test rationality on that variation. This is GARP-conditional-on- Z , and Chapters 2–4 each instantiate it.

1.5.1 The ACD goodness-of-fit statistic, concretely

For a model that predicts changes Δx_n in outcomes n , and observed changes Δy_n , ACD’s statistic with instrument z_n (a shift-share of exogenous shifters) is

$$\hat{\beta}_z = \frac{1}{N} \sum_n z_n (\Delta y_n - \Delta x_n).$$

Under the null that the model’s causal prediction is correct, and with z orthogonal to the nuisance, $\mathbb{E}[\hat{\beta}_z] = 0$; its magnitude estimates the average misspecification. The same object, with “prediction error” replaced by “revealed-preference residual,” is what you test in the choice setting.

1.5.2 Binning: a practical way to test “conditional on Z ”

Intuition — Testing restrictions over bins of Z

Often you do not want a single moment; you want the theory to hold *at every value* of Z (a conditional restriction). The workhorse trick is **binning**: partition observations into groups defined by Z (e.g. states by legalization status, periods by announced-tax timing, prompts by frame), and (i) test the revealed-preference / moment inequalities *within* each bin, and/or (ii) test that the theory’s key object is *stable across* bins. Conditioning on Z in bins converts a conditional-moment-inequality problem into a finite collection of unconditional ones — which is exactly the structure the inference tool below handles. (This is the “over bins” device from your GARP work, generalized.)

1.6 Inference: testing inequalities correctly, and power

Why you need moment-inequality inference (Andrews–Roth–Pakes 2023)

Deductive theories imply inequalities, so ordinary regression p -values do not apply. ARP (2023) give tests of *linear conditional moment inequalities* with correct size even with many inequalities and nuisance parameters — precisely the RP/Afriat setting. Crucially, most of the prior empirical literature reports an *index* (CCEI, an efficiency score) with *no* sampling theory, so it cannot say whether a deviation is real or chance. **Bringing correct inequality inference to a setting that lacked it is itself a contribution.**

Two cautions: power and noise

Power. A test nothing can fail is useless. Random choosers violate GARP almost always (the **Bronars 1987** benchmark: simulate uniform-random choices and see how often they fail), so “passing GARP” is informative only against that baseline. Crawford (2010) warns that some RP tests have *low* power unless the price variation is rich — another reason Z matters: it supplies powerful, exogenous variation.

Noise. Real choices, and an AI’s stochastic decoding, contain randomness. **Statistical / stochastic revealed preference** (Chambers–Echenique 2016; Varian 1985 for measurement error) plus ARP inference separate *systematic* violations from chance. A raw CCEI cannot.

1.7 Welfare: why the verdict matters (Bernheim–Rangel 2009)

Welfare without assuming rationality

If the robust test says choices are inconsistent (they flip with framing or timing), what is “welfare”? Bernheim–Rangel replace revealed preference with the **unambiguous choice relation** P^* : $x P^* y$ iff y is *never* chosen when x is available, in a **welfare-relevant domain** that prunes “**ancillary conditions**” — framing, timing, salience, anchors — that affect choice but that a planner need not honour. When choices conflict only across ancillary conditions, welfare is *ambiguous*, and that ambiguity is the finding. This yields generalized consumer surplus, compensating variation, and welfare theorems, with standard analysis as

the limiting case when anomalies are small.

Intuition — “What is welfare?” has rival answers

Decision utility (what choice reveals; Bernheim–Rangel respect this) vs. **experienced utility** (hedonic; Kahneman) vs. **remembered utility** (recalled, and distorted — Kahneman’s “two selves,” peak-end, duration neglect). A choice-based test diagnoses *whether* stable preferences exist; it does not settle *which self’s* welfare counts. Be explicit about that scope — it is a strength.

1.8 The recipe, and how the rest of the book uses it

One recipe, three applications

deductive theory $\Rightarrow$ RP/moment inequalities $\Rightarrow$ confront data through $Z \perp$ nuisance $\Rightarrow$ ARP inference $\Rightarrow$ Type A/B verdict $\Rightarrow$ welfare/policy.

Chapters 2–4 each walk this path: the theory and its behavioral rival; the predictions as inequalities; the existing tests and what they assume; the precise gap; and a concrete new test (the Z , the binning, the inference, the data, and what would falsify it).

Essential readings — Chapter 1 — method

Read in this order. **(1)** Echenique (2020, *Ann. Rev. Econ.*) — the RP ladder, short and unifying. **(2)** Chambers & Echenique (2016), *Revealed Preference Theory*, chs. 1–3 and the statistical-RP chapter — the rigorous backbone. **(3)** Adão–Costinot–Donaldson (2025, *QJE*) §I–II — the two-forces logic and the IV test. **(4)** Andrews–Roth–Pakes (2023, *REStud*) — skim the setup and the main theorem for moment-inequality inference. **(5)** Bernheim & Rangel (2009, *QJE*) §I–II — ancillary conditions and the unambiguous choice relation. **(6)** Afriat (1967) / Varian (1982) — the original GARP theorem (or get them via Chambers–Echenique).

Chapter 2

Rationality over Time: How Much Present Bias Is Real, and When?

2.1 The question

Ask an economist why people under-save, over-eat, procrastinate, smoke, scroll, or hit snooze, and the standard answer is *present bias*: a systematic over-weighting of the present that makes today's self sabotage the plans of yesterday's self. The concept anchors a vast policy apparatus — sin taxes, default enrolment, commitment devices, screen-time limits. Yet the empirical foundation is shakier than the apparatus suggests, and in a specific, fixable way.

The one idea this chapter is built on

“Present bias” as usually measured is not one thing; it is a *composite*. When we observe someone choose impatiently now, what we have measured is a sum: a genuine self-control problem, *plus* the shadow of implicit risk, liquidity constraints, the fungibility of money, measurement error, household bargaining, and mistaken forecasts of one's own future tastes. Each of those can produce the exact behavioural signature of present bias *with no self-control problem at all*. The chapter's job is to *decompose* the composite — to ask how much of it is genuine — and then to ask a second question the literature has barely posed: whether what survives is a fixed personal *trait* or a *state* that the environment switches on and off.

Those two questions — **how much is genuine**, and **is it a trait or a state** — are not academic. They carry opposite policy prescriptions. If most measured present bias is really liquidity, the remedy is credit, not commitment. If it is a stable trait, you tag the person (a tax, a generic default). If it is a state manufactured by cues and cognitive load, you can also manage the *environment* that triggers it. And if a given behaviour is one a person would endorse on reflection, the right amount of paternalism is *none*. A credible decomposition, with credible identification, is what lets policy tell these cases apart instead of treating every impatient choice as a defect to be corrected.

Intuition — The contribution in one sentence

Take the behaviour everyone calls “present bias,” run it through a *system* of switches that turn off each impostor in turn — make the future certain, loosen the budget, use effort not money, handle noise, keep the decision individual — and see what is left; then move the person's *state* (proximity, cue, load, visceral heat) and ask whether the residual is flat (a trait) or sloped (a reaction function). What survives, and how it behaves, is the object policy should actually target.

2.2 What we already know — and the turn that reframes the question

Three results define the terrain. Taking them seriously is what separates this project from “re-measuring present bias.”

(i) **It shows up over effort, not money — but the real trigger is immediacy.** Andreoni–Sprenger (2012a) and Augenblick–Niederle–Sprenger (2015) find little present bias when people allocate *money* over time, but substantial present bias when they allocate *effort* (unpleasant tasks). The natural reading — “money is fungible, so people arbitrage it; effort is consumed when done, so it reveals true impatience” — is only half right. Balakrishnan, Haushofer & Jakiela (2020) show that money present bias *reappears* ($\beta \approx 0.90$) once the same-day payment is made *truly immediate* rather than arriving hours later. So the effort–money contrast is, at least in part, an *immediacy* contrast (Weber–Chapman 2005 and Keren–Roelofsma 1995: adding delay dissolves the certainty effect and adding risk dissolves the immediacy effect — the two are entangled). Lesson: the domain label is not the fundamental; *true immediacy of the consumed good* is.

(ii) **Time-inconsistency is not the same as present bias (Halevy 2015).** The standard “static” present-bias test compares a choice made today with a choice about two future dates; Halevy proves it identifies only *stationarity*, and that inferring genuine time-inconsistency requires a separate, almost-never-tested assumption (*time-invariance*). Empirically he finds roughly half of “time-inconsistent” people are not present-biased at all. Whatever the residual is, a single snapshot cannot see it: you need choices from *multiple vantage points*.

The 2021 turn: noise and aggregation overturn the flagship field evidence

Two recent results change what the chapter must explain. **Measurement error alone reverses the survey verdict.** Aguiar–Kashaev (2021) build the first fully nonparametric *statistical* revealed-preference test that allows for measurement error, and apply it to the Spanish expenditure panel (ECPF) that the nonparametric present-bias literature was built on. The deterministic test rejects exponential discounting for **84% of single-person households**; once trembling-hand noise is handled, exponential discounting **cannot be rejected for singles at all** (even with the discount factor set to one). In their Monte Carlo the deterministic test falsely rejects a true exponential model 79% of the time. **And what looks like household present bias is largely aggregation.** Adams et al. (2014) show that letting spouses have *different but constant* discount rates rationalises 97% of couples; Jackson–Yariv (2014, 2015) prove that aggregating any heterogeneous exponential discounters *necessarily* produces present-biased-looking group choices (Millner–Heal 2018 sharpen this: it is the collective analogue of Halevy’s stationarity/invariance/consistency distinction). So the field evidence for individual present bias in consumption is, on the frontier’s own methods, dissolving.

This is the opening. The literature stands on evidence its own best tools are starting to overturn; the honest questions — how much genuine present bias is left, and is it a trait or a state — are exactly the ones no one has answered with credible identification.

2.3 Seven impostors, and an honest map of what is separable

Write the object of study as a *wedge*: for person i , let r_i be the difference between what she *planned* to do (deciding ahead) and what she *actually did* (on the day), for the same dated action. A negative wedge — doing less of an unpleasant task now than planned — is the raw behavioural signature of present bias. The whole difficulty is that seven distinct things produce $r_i < 0$, only one of which is a self-control problem.

The seven channels that can each produce a “present-bias” wedge

1. Genuine present bias.

A real conflict between today’s self and the plan (quasi-hyperbolic $\beta < 1$, or a dual-self willpower cost). *The only channel that is a self-control problem.*

2. Implicit risk.

A future reward is uncertain; a person who is merely risk-sensitive (or weights probabilities) discounts it more, which looks like impatience (Halevy 2008; Epper–Fehr–Duda–Bruhin 2011).

3. Liquidity / income expectations.

A credit-constrained person who expects more cash later rationally wants resources sooner (Dean–Sautmann 2021; Cassidy 2019 — causal; Epper 2017 — expectations).

4. Money arbitrage / immediacy.

Money is fungible, so monetary “impatience” reflects outside interest rates, not preferences; and the residual is an *immediacy* effect, not a domain effect (ANS 2015; Balakrishnan et al. 2020).

5. Measurement error.

Noisy choices make deterministic consistency tests over-reject a rational model (Aguiar–Kashaev 2021).

6. Household aggregation.

Heterogeneous exponential members produce present-biased *group* choices with no individual bias (Adams et al. 2014; Jackson–Yariv).

7. Misprediction / projection.

A person mis-forecasts her own future tastes (from her current state), so the plan was wrong, not the will (Loewenstein–O’Donoghue–Rabin 2003; Augenblick–Rabin 2019; Badger et al. 2007). And *myopia* (Gabaix–Laibson cognitive noise) produces present-focus with underlying-consistent preferences and *no* commitment demand.

Channels 2–6 are not self-control problems at all; channel 7 is a genuine bias but a *different* one, with a different remedy (information/design, not commitment). Only channel 1 warrants the classic self-control policy.

Intuition — A running example: Anna’s twelve gym visits

Keep one concrete person in mind throughout. In January Anna plans twelve gym visits for February; February delivers five. Before calling her present-biased, walk the channels: perhaps she rationally hedged an uncertain month (*risk*); perhaps a tight paycheck made the \$5 day-pass bite (*liquidity*); perhaps her utility is simply concave so the plan and the action are both fine and the *scale* of the gap misleads (*curvature*); perhaps getting out the door has a fixed cost that no per-visit reasoning captures (*fixed cost/temptation*); perhaps she planned while energized after a good week (*projection*); perhaps she remembers going eight times last month when the turnstile counted five, so the plan was anchored on a false past (*memory* — in the Carrera data this inflation is +8.4 visits per hundred days, measured directly); perhaps the “she” who planned was a couple’s joint calendar (*aggregation*); and perhaps the count itself is noisy (*measurement*). Each channel has a different policy remedy, and each has an observable *switch*: something that turns it on or off while leaving the others alone (Figure 2.1). The chapter’s whole method is: build the wedge, then run the switches, then say — with set-identified shares, not bravado — how much of the population’s wedge survives every switch. That surviving part is the object policy cares about.

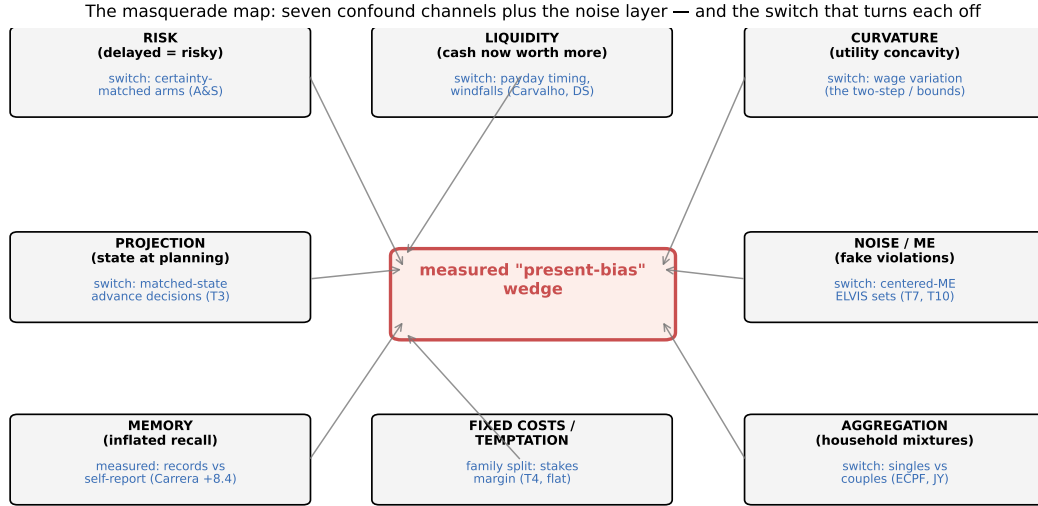


Figure 2.1: The masquerade map. Every box is a mechanism that can generate Anna’s gap without any self-control failure (plus the noise layer, which manufactures violations out of nothing); the blue line inside each box is the switch — an instrument Z or a design contrast — that turns that channel off. The tests in Section 2.14 are these switches, executed.

The honest identification map (Ericson–Laibson 2019): what is separable from what

These accounts are *observationally close* on any single margin. Quasi-hyperbolic collapses to exponential as $\beta \rightarrow 1$ (Aguiar–Kashaev’s own caveat); exponential and hyperbolic both fit the same money data (Andersen et al. 2008); a *fixed cost* of acting now fits as well as a multiplicative β (Benhabib–Bisin–Schotter 2010); cognitive-noise myopia mimics present bias with no self-control problem. This is not a defect to hide; it is the reason the chapter’s object is a **decomposition with set-identified shares and a nested horse race**, not a claim to point-identify each person’s model. Ericson & Laibson (2019) give the map that keeps this disciplined: present-focused models sort on a 2×2 of **(taste for commitment: yes/no) $\times$ (dynamic consistency: consistent/inconsistent)**. Two facts from that map drive everything below: *commitment demand is the observable “smoking gun” for a self-control problem* — but it is confounded (even a sophisticate declines commitment when the future is uncertain), so it must be triangulated, never trusted alone; and *dynamic consistency is not measurable from binding choices — only by eliciting state-contingent preferences across vantages*, which is exactly the multi-vantage plus prediction data this design collects.

2.4 The models on trial (this is not a test of quasi-hyperbolic discounting)

A recurring misreading should be headed off at once: the paper does not fit, or test, the quasi-hyperbolic model. It runs a *nonparametric horse race* over a ladder of models, of which quasi-hyperbolic is one rung.

(E) Exponential discounting

— the rational, time-consistent benchmark. $\sum_{\tau} \delta^{\tau} u(e_{\tau})$. Never wants commitment.

(QH) Quasi-hyperbolic

- one extra factor $\beta < 1$ on all future dates (Laibson 1997). The reference *behavioural* model. Nests (E) at $\beta = 1$.

(SQH) State-dependent quasi-hyperbolic

- β is a function $\beta_{\text{eff}}(X)$ of the state, not a constant. Nests (QH) as the flat special case. *The chapter's new object.*

(PD) Planner–doer / dual-self

- costly willpower (Fudenberg–Levine 2006); predicts the wedge *shrinks* with stakes and co-moves with small-stakes risk aversion.

(T) Temptation

- a smaller menu can be strictly preferred (Gul–Pesendorfer 2001); commitment demand *without* a preference reversal.

(FC) Fixed-cost present bias

- a fixed utility cost of acting now, not a proportional β (Benhabib–Bisin–Schotter 2010); the wedge is roughly constant in absolute terms, so its *share* shrinks with stakes.

(M) Myopia

- forecasts shaded toward a prior (Gabaix–Laibson); present-focus, underlying-consistent, *no* commitment.

(A) Aggregation

- the “person” is really several (household, or several selves); heterogeneous exponentials look present-biased in aggregate.

The models are compared nonparametrically, using revealed-preference *inequalities* (no functional form) and a predictive-success criterion (Beatty–Crawford 2011) that corrects for how permissive each model is, so that “QH fits better” cannot be an artefact of QH’s extra freedom. Where models are nested ($E \subset QH \subset SQH$) the comparison is a direct inequality test; where they are not (PD vs QH; FC vs QH) the discriminators are the stakes response, the risk link, and the shape of the wedge.

One more axis the models must respect: the four sign-timing configurations of a choice

“Intertemporal choice” bundles four structurally different trade-offs (Figure 2.2), and the domains of this book scatter across all of them: **(R–R)** a reward now vs. a reward later (money MELs and CTBs; Sun–Potters); **(C–C)** a cost now vs. a cost later (the effort panels: A&R, ANS, Fedyk — postpone the unpleasant tasks or not); **(R⁺C[−])** a pleasure now with a cost later (smoking, scrolling, betting, the snack — the classic temptation cell: AGS, Sadoff); and **(C⁺R[−])** a cost now with a reward later (exercise, studying, saving: StepUp, OULAD, SEED). In the pure discounting models these four are the *same* object — only the dates of the utils matter, so one (β, δ) prices them all, and that symmetry is itself a testable restriction. The data break it in exactly the ways the chapter documents: the money wedge is *reverse-signed* where the effort wedge is positive (ANS); anticipation makes some people do painful things *early* to stop dreading them (Test B; Falk–Zimmermann) — impossible in any (β, δ) model where a cost’s date only makes it lighter; and domain coherence fails ($\beta_e \neq \beta_m$). The MSC scaffold absorbs this without new machinery, because in MSC the distortion attaches not to the *date* but to the *immediate, cue-laden component* of the choice, whatever its sign: temptation inflates immediate pleasures (the R⁺C[−] cell), aversion-plus-dread inflates immediate pains (the C–C and C⁺R[−] cells, with anticipation pulling the other way), and pure money trade-offs, having no visceral immediate component, show

little genuine bias at all — which is what the field consumption data and Sun–Potters find. So the answer to “should the models change?” is: the *parametric* models cannot express the asymmetry and are tested against it; MSC already predicts it, and the taxonomy’s role is to make the prediction explicit and to stop results from different cells being averaged as if they measured one number. And a cell is not a single object either: each trade-off carries a *size* and a *date* for *each* of its two components, and a pure (β, δ) prices all of these symmetrically — only the dates enter, so the size of the reward and the size of the cost, near date and far date, are all treated alike. MSC is not symmetric here: because the distortion rides the *immediate, cue-laden* component, it should scale with *that* component’s magnitude and salience and be largely insensitive to the *delayed* component’s size. The stakes and domain results already probe this finer prediction — the bias sits on effort (the immediate cost) and barely tracks the delayed reward $(+.07/\log\text{-wage})$ — and a design that varies the two magnitudes independently would settle it (the sixth row of Table 2.1). Results from different cells are never averaged as one number in this chapter: the effort sets, the money wedges, the temptation-goods wedges, and the investment-goods programs are reported cell by cell.

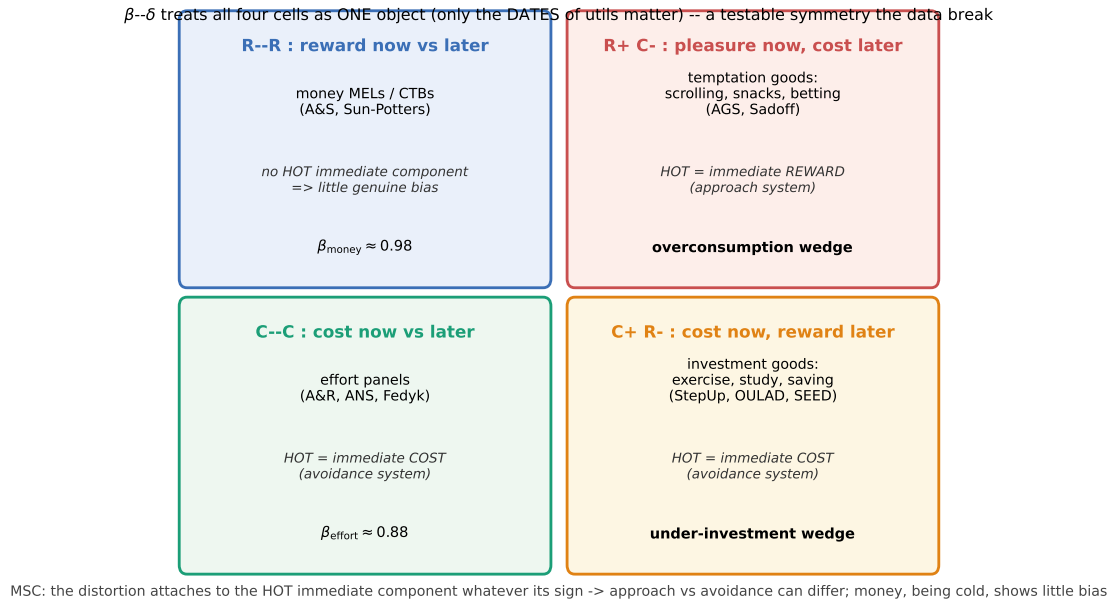


Figure 2.2: The taxonomy of intertemporal trade-offs. The two rows are the sign of the *immediate* component (reward vs. cost); within each cell the delayed component has the opposite sign for temptation and investment goods. A pure (β, δ) model prices all four cells with the same present-bias factor (only the dates of the utils enter); MSC attaches the distortion to the *hot immediate component*, so approach (immediate reward) and avoidance (immediate cost) can differ, and money — with no visceral immediate component — shows little bias. Each cell names the datasets that populate it.

This turns a scattered list of discounting anomalies into a single, organised *symmetry audit*. A (β, δ) model applies one present-bias factor to $\delta^t u(x_t)$ regardless of the util’s sign, size, domain, or the date from which it is viewed. Each of those invariances is a restriction, and the taxonomy says exactly which contrast tests it (Table 2.1). The chapter’s organising empirical claim is that *four of the five restrictions are rejected*, each by its own contrast, and that MSC breaks precisely these and no others.

Table 2.1: The β - δ symmetry audit: each invariance, the taxonomy contrast that tests it, and the executed verdict.

Invariance (β, δ) imposes	Taxonomy contrast that tests it	Verdict (executed)
Domain/system: one β for money and effort	R-R money vs. C-C effort, <i>same people</i> (ANS 2015)	rejected: $\beta_{\text{eff}} = .88$ vs. $\beta_{\text{money}} = .99$, gap $+.10$ $ [.04, .19]$, $p < .001$
Magnitude: $\beta \perp$ stake size	β by wage quintile (A&R)	modestly rejected: exact flatness $p = .001$; trend $+.07/\log\text{-wage}$ toward 1
Future stationarity: no drift among future dates	proximity gradient (A&R, OULAD, KKM)	rejected: graded gradient, CS/ARP $p < .001$
Separability: a cost’s date only lightens it (no anticipation)	do-it-early / dread (Test B; Falk–Zimmermann, Ganguly–Tasoff)	rejected: anticipation reverses the sign for some pains
Component magnitude: β responds equally to the size of the <i>immediate</i> vs the <i>delayed</i> side	size of the hot immediate component (ASYM) vs the delayed reward (STK piece rate)	partially rejected: the bias loads on the <i>hot</i> side — it lives on effort (immediate cost) not money, and tracks the delayed reward only weakly ($+.07/\log\text{-wage}$)
Sign <i>within</i> domain: $\beta(\text{gain}) = \beta(\text{loss})$	gain vs. loss framing of the <i>same</i> good	<i>not separately identified here</i> (money–effort confounds domain with sign) — flagged for the sequel

2.5 The design: a wedge over effort, from many vantages, with switches

The measurement instrument is the *convex time budget* over effort (Andreoni–Sprenger 2012a; ANS 2015). Instead of a yes/no choice, a person splits a fixed block of unpleasant tasks between an earlier and a later work date at an experimental exchange rate R (“every task postponed costs R later tasks”). Sliding the split as R varies traces a smooth demand for effort-now-vs-later, which pins down both impatience and cost curvature — information a binary choice cannot give.

Intuition — What identifies present bias here is *dynamic*, not the level

Effort is a cost, so “impatience” means *procrastinating* — pushing work later. But the level of procrastination is not the test; a patient exponential type also has a preferred split. Present bias is identified by *comparing the plan to the realisation for the same dates*: when the earlier date *becomes today*, a present-biased person does less now than she had planned, so the wedge $r = e_{\text{early}}^{\text{real}} - e_{\text{early}}^{\text{plan}} < 0$. Because this compares a plan with its own execution, it tests *time consistency* (the welfare-relevant property), not mere stationarity — Halevy’s distinction, built into the instrument.

Two design features do the identifying work. **Multiple vantage points:** each person is observed deciding about the same dated work from several distances, so consistency (not stationarity) is what is measured. **Switches Z :** conditions that turn a suspected impostor on or off, so that differencing the wedge across them attributes it to its source. Figure 2.3 shows where each object lives on the time line — and why the advance vantage is special.

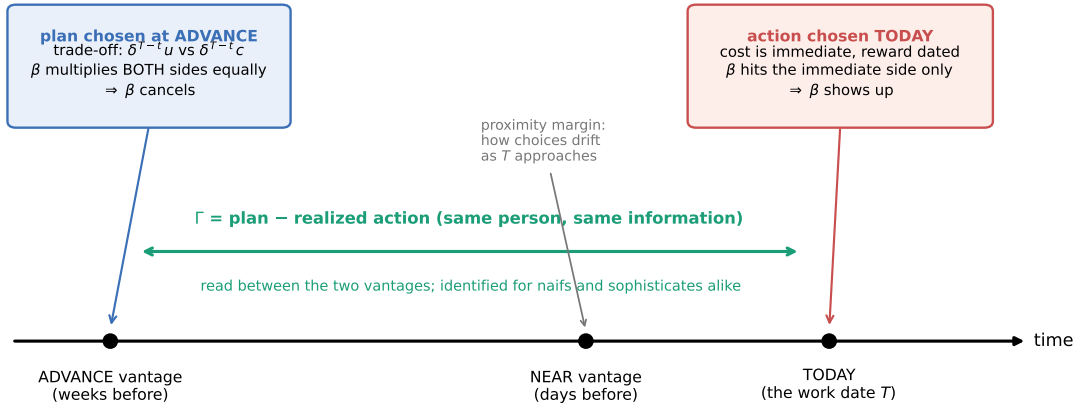


Figure 2.3: The vantage timeline. At the *advance* vantage both the work date and its reward are in the future, so any present-bias factor β multiplies both sides of the trade-off and cancels: advance choices reveal the long-run preferences (δ , cost curvature) that every “self” shares. At *today*, the cost is immediate and β hits one side only, so it shows up. The welfare wedge Γ is the difference between the two — the same person disagreeing with herself, with no outside standard imported. The near vantage in between traces *how* the disagreement builds as the date approaches (the proximity margin).

Intuition — The proximity margin is not present bias itself — it is a test OF the model of present bias

Distinguish two questions the timeline separates. *Does the wedge exist?* — compare advance and today; that is the present-bias test. *How does choice change across future vantages, before anything is today?* — that is the proximity margin, and the textbook model makes a stark prediction about it: quasi-hyperbolic β is a *cliff at now*. All strictly future dates are discounted exponentially, so a QH agent deciding at 15 days out and again at 5 days out (both still future) should not budge. A *graded* drift — allocations tilting away from the work date already at 4–7 days (A&R, GMS $p = .007$), study effort doubling inside the last two days (OULAD), output climbing into payday (KKM) — says the “present” is a *zone*, not a point: the discount weight is a declining function of distance, as visceral-anticipation and attention accounts predict and as literal $\beta\text{--}\delta$ does not. Practically, the proximity slope is where deadlines, reminders, and cooling-off periods get their leverage; theoretically, it is one of the sharpest wedges *between* behavioural models, which is why the chapter reports it separately from the existence result and never conflates the two. Two more questions the timeline settles. *Could the gradient just be δ (or β)?* No: a choice between effort at T_1 and T_2 , made at decision time t , trades off δ^{T_1-t} against δ^{T_2-t} , and the ratio $\delta^{T_1-T_2}$ does not contain t — *any* stationary discounter, quasi-hyperbolic included while both dates remain future, chooses the same split from every distance. The observed drift is therefore on top of β and δ , not a rediscovery of them. *Could it be uncertainty?* Partly, and honestly so: far dates are blurrier, and as information arrives an option-value motive can shift allocations (Halevy 2008 ties impatience to hazard). The lab version is disciplined by the certainty-matched arms; the field execution profiles (OULAD clicks, KKM output) additionally mix in *rational* scheduling — deadlines coordinate work — which is why the within-person allocation drift on A&R is the sharp exhibit and the field gradients are corroboration at scale, not stand-alone proof.

Is the “switch” idea actually new? The honest audit

Each individual switch has an ancestor, and none of the ancestors adds up to the design. Andreoni–Sprenger (2012b) *did* randomise payment risk — over money, with no immediacy, in a static design, as a common-ratio check, with no decomposition. Balakrishnan et al. (2020) randomised immediacy — one switch, money only. Carvalho et al. (2016) and Dean–Sautmann (2021) exploit *quasi*-random payday/shock timing — correlational, money. ANS (2015) and Augenblick–Rabin (2019) built the effort panels — but *minimise* confounds by protocol (“passive cleaning”: pay in cash, build trust) rather than *randomising* them (“active cleaning”), so when they find effort present bias they cannot show it is not setting-specific risk or time-liquidity. What no one has is the *system*: a within-person battery of switches over *effort with multiple vantages*, turning “a wedge exists” into “the wedge decomposes, person by person,” plus formal inference on the shares, plus the trait-vs-state slope, plus a noise layer so none of it is an artefact of measurement error. Claim the system; credit the ancestors for the parts.

2.5.1 Can we actually build these switches from data we can reuse? (the feasibility audit)

This is the decisive practical question, because the plan is to *reuse existing experimental and field data*, not to run a new experiment. A switch is only real if the data contain genuine (quasi-)random variation in it. The honest audit:

Switch	Randomised variation in data on hand	Verdict
Z_{risk}	A&S 2012b (randomised certain vs risky, common-ratio); Ganguly–Tasoff 2016	Available (money; experimental).
Z_{liq}	Carvalho 2016 (randomised payday timing); Cassidy 2019 (randomised windfalls); Dean–Sautmann 2021 (within-person income shocks)	Available (Carvalho/Cassidy truly randomised; D–S quasi).
$Z_{\text{immediacy/prox}}$	ANS 2015 / A&R 2019 (design-varied distance-to-work); Augenblick 2018 (within-week); Giné et al. 2016 (randomised closeness)	Available (proximity), though the fine near-region is sparse in A&R.
Z_{visc}	Schilbach 2019 (randomised sobriety incentives); Badger 2007 (craving before/after dose)	Partly (Schilbach randomised; Badger quasi).
Z_{stakes}	A&R 2019 / ANS 2015 (wage grid); A&S (amounts)	Available (experimental).
Z_{dom}	ANS 2015 (effort and money in-subject)	Available.
$Z_{\text{cue}}, Z_{\text{load}}$	<i>No open dataset randomises a consumption cue or cognitive load crossed with an intertemporal choice</i> (Deck–Jahedi 2015 inaccessible; Carvalho has executive <i>levels</i> , not randomised load)	Gap.

What this means for the paper — stated plainly

Five of the switches (risk, liquidity, immediacy/proximity, stakes, domain) and one visceral proxy are *buildable now* from randomised or quasi-random variation in datasets already in hand. So the *decomposition* of the wedge into genuine-vs-confound, and the trait-vs-state test on the *proximity, stakes, and visceral* margins, are executable on real data without any new experiment. The one margin without a purpose-built dataset is high-dose randomised *cue/load*. Even here the paper is not silent: the *load* half is tested on the doses open data do supply — measured-sleep bins and randomized cognitive load both return bounded nulls (Section 2.14.3) — so what is genuinely missing is a *high-dose, purpose-built cue* manipulation. Three honest options follow: **(a)** run trait-vs-state on every margin open data *do* supply (proximity, stakes, visceral, liquidity-as-moderator, and the executive/load channel), already a strong multi-channel result; **(b)** seed the pure-cue cell with a ≤ 5 -minute free micro-survey (hypothetical choices with within-person digit-load and immediacy randomised — flagged exploratory); **(c)** state the clean high-dose cue experiment as the pre-registered prediction. The paper leads with (a), treats (b) as a cheap seed, and (c) as honest scoping; it does *not* promise a result it cannot deliver from existing data.

2.6 The formal test

2.6.1 Step 1 — a consistency screen (not the present-bias test)

Within each vantage and each Z -cell, the allocations across budgets $\{R\}$ must be rationalisable by *some* stable within-period preference — they must satisfy GARP / the Afriat inequalities of Chapter 1. This is a data-quality gate; it is silent about time or present bias. The present-bias content lives in the *across-date* conditions and in the wedge, not here.

2.6.2 Step 2 — the noise layer, and the division of labour between the two engines

Every deterministic revealed-preference verdict inherits the Aguiar–Kashaev lesson: with noisy choices, exact inequality checks over-reject (79% false rejection of a true exponential model in their Monte Carlo). The fix is to run the inequalities as *statistical* hypotheses that integrate out both measurement error and unobserved heterogeneity in (β, δ, u) .

The ELVIS layer, in one paragraph

Write the model’s Afriat-type inequalities for *true* quantities = observed – error, and add what is *known* about the noise as a *centering condition* (experiments: trembling-hand $\mathbb{E}[w] = 0$, validated on Kurtz-David et al.’s 2019 target-vs-click data; surveys: mean-budget neutrality $\mathbb{E}[\rho'w] = 0$). The model is consistent with the data iff a distribution of latent (β, δ, u, w) exists satisfying the inequalities almost surely plus the centering — an existence problem that Schennach’s (2014) *entropic latent-variable integration* (ELVIS) turns into a finite-dimensional simulation with a χ^2 statistic. Two properties matter: it **drops nobody** (contrast the structural approach of Augenblick–Rabin, which discards 28% of subjects its cost function cannot rationalise — ELVIS integrates over *all* rationalisable stories per subject, with unrestricted heterogeneity), and it **accepts instrument moments** (Aguiar–Kashaev §6.4 allow orthogonality conditions $\mathbb{E}[z'w] = 0$), so the randomised Z ’s enter the noise layer natively.

Two engines, two jobs — and why selection is *not* a problem here

Keep the layers straight. **ELVIS (model layer)** answers: does *some* model in the ladder (E / QH / SQH / A) rationalise the noisy panel, integrating out error and heterogeneity? It delivers model verdicts and set-identified parameter bounds (e.g. the support of β), *not* per-person point estimates. **ARP / Cox–Shi (design layer)** answers the reduced-form questions — the wedge decomposition, the flatness/ordering of β_{eff} , the type-share bounds — via conditional moment inequalities on the wedges themselves. *On the selection worry*: in earlier revealed-preference work a moment-inequality test could induce selection, because it conditioned on a subject’s *history* being rationalisable (keeping only GARP-consistent references is selection on the outcome). That construction does not appear here: the design-layer tests condition on nothing — every subject contributes her wedge across randomised Z -cells, and mean-zero error plus randomisation keep $\mathbb{E}[r \mid Z]$ consistent. So ARP/Cox–Shi run cleanly on the full sample. ELVIS is reserved for the model layer, where its value (no dropping under measurement error) is real; it can also, if desired, absorb the confound and Z moments into a single conditional test $\mathbb{E}[g \mid Z] = 0$ (Schennach’s conditional-moment extension — more powerful than $\mathbb{E}[w] = 0$ or $\mathbb{E}[z'w] = 0$, at higher computational cost). The two engines are complements: ELVIS guards the model verdicts against noise; ARP guards the design contrasts at the boundary.

2.6.3 Step 3 — the decomposition and the population shares

Let r_i be person i 's planned-minus-realised wedge. The rational model says that, *with the confounds switched off*, there is no gap:

$$\mathbb{E}[r_i \mid Z_{\text{risk}} = \text{certain}, Z_{\text{liq}} = \text{slack}, Z_{\text{dom}} = \text{effort/immediate}] = 0.$$

Present bias predicts $r_i < 0$ *even in that clean cell*; each confound predicts $r_i < 0$ *only* where its switch is on. Differencing across cells attributes the wedge to its sources, and the population *shares* of the types — {exponential, genuine-PB, risk-driven, liquidity-driven, money/immediacy, aggregated, misprediction} — are *set-identified* (a linear program over the simplex of shares against the confidence box on the observed cell frequencies), with Andrews–Roth–Pakes / Cox–Shi inference. The headline object is a bound: “genuine present bias is between $x\%$ and $y\%$ of the measured inconsistency,” not a point estimate.

2.6.4 Step 4 — trait vs state, as a revealed-preference characterisation

The flagship test asks whether the cleaned wedge is a constant $\bar{\beta}$ (trait) or a reaction function $\beta_{\text{eff}}(X)$ that falls as the state heats up. Blow–Browning–Crawford close their paper by conjecturing that revealed-preference conditions for *other* discounting models should exist; here is the one for this chapter.

RP characterisation of state-dependent quasi-hyperbolic discounting

The model. In state cell c the agent weights dates by $w(0) = 1$, $w(k) = \beta(c)\delta^k$ ($k > 0$), a single δ , and $\beta : \mathcal{C} \rightarrow (0, 1]$ constant *within* a cell, free *across* cells except for monotonicity ($\beta(c) \geq \beta(c')$ when c is cooler). **The conditions.** (i) Within each cell, the choices satisfy the Blow–Browning–Crawford quasi-hyperbolic Afriat conditions with parameters (β_c, δ) — a finite linear-in-unknowns system whose feasible (β_c, δ) region B_c is computable, and which *bounds* β_c . (ii) Across cells, a common δ and a monotone selection $\beta_c \in B_c$ exist — a linear feasibility problem on the intersection of the per-cell bound sets. **Nesting.** The trait model is $\beta_c \equiv \bar{\beta}$: its condition is $\bigcap_c B_c \neq \emptyset$. So “trait vs state” is a nested pair of feasibility systems, and the fingerprint of state-dependence is precisely a set of cells whose β -bounds are pairwise disjoint *in the monotone order*. **Statistics.** Deterministically each layer is a linear program; statistically, the within-cell conditions run through ELVIS (state-dependence is a per-cell restriction on the latent supports) and the cross-cell monotone selection is a moment-inequality system for ARP / Cox–Shi. Randomisation is what makes it clean: because cells are assigned exogenously, disjoint β -bounds across cells cannot be blamed on selection into states.

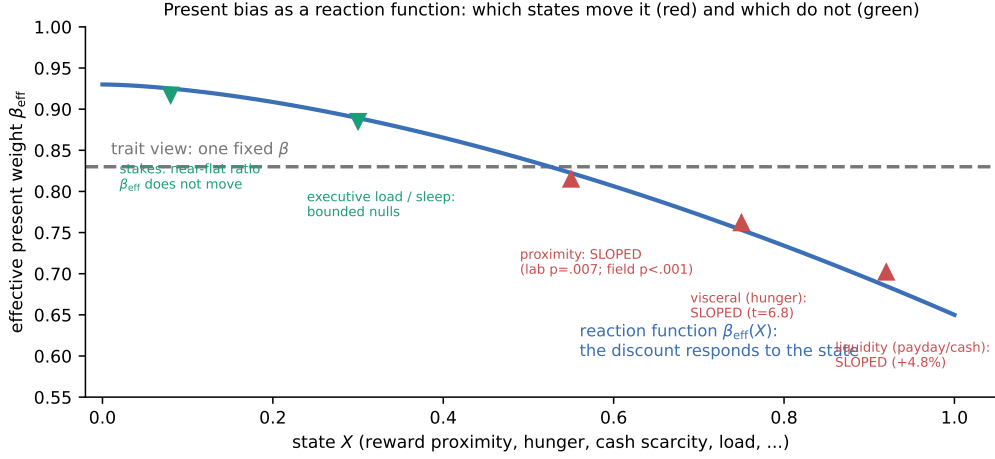


Figure 2.4: The chapter’s new object, schematically. The trait view (grey dashed) says one number describes the person; the reaction-function view (blue) says the effective present weight responds to the state. The markers preview the executed verdicts of Section 2.14: the margins that barely move β_{eff} (stakes: at most a modest rise toward 1; executive load and sleep: bounded nulls at tested doses) and the margins that *do* (reward proximity, visceral hunger, cash scarcity). “Present bias” is neither one number nor infinitely labile — it is a low-dimensional reaction function with a specific, replicated shape.

Intuition — The scale-free two-step, with numbers: why wage variation is the exchange rate

Here is the whole trick in one worked example. Suppose Anna does $e = 100$ tasks when the piece rate is $w = 10$ cents and 121 tasks at $w = 20$ cents. That response measures her *cost curvature*: a doubling of the value of work raised her effort by $\log(121/100) \approx .19$ log points, so $b \equiv \partial \log e / \partial \log w \approx .28$. Read b as an *exchange rate*: it says how many log-points of *effort* one log-point of *value* buys, for Anna, on this task. Now observe her present-bias wedge: when the work is *today* rather than dated, she does $\theta = -.10$ log points less at the same wage. Since β enters exactly like a devaluation of the (dated) reward, the wedge in value units is $\theta/b \approx -.36$, i.e. $\hat{\beta} = e^{\theta/b} \approx .70$. No utility scale, no task units, no interpersonal comparability was used — both θ and b are within-person log responses, and their ratio is dimensionless: that is what “scale-free” means. Two corollaries the results exploit. *First*, the raw tasks-wedge is a lie detector’s nightmare: with convex costs it grows mechanically with stakes even under constant β , so state-dependence must never be read off task counts (the pilot’s Lesson 1). *Second*, the two-step lives and dies with the wage response: if the effort–wage curve is nearly flat ($b \approx 0$, as in DellaVigna–Pope’s real-effort task, $b = .03$), the division by b explodes and $\hat{\beta}$ is garbage — our estimate there is a meaningless 1.16 — while the *matching bound* stays sane: find the wage w^* at which her dated-work effort equals her today-effort at w ; monotonicity alone gives $\beta = w^*/w$ as a bracket, with no cost function at all. That division of labour — point estimate where the wage curve has slope, bounds where it does not, ME-robust sets everywhere — is the three-rung ladder of the results section.

2.7 Two measurement lessons the effort panel teaches

Before the results (Section 2.14), two lessons from the first pass over the Augenblick–Rabin panel (11,405 decisions, wage grid, choices at varying distance-to-work) that now discipline every estimate in this chapter.

Lesson 1: the level-wedge lies; measure scale-free or in bounds

The raw *level* wedge (tasks foregone when work is immediate) appears strongly stake-dependent — it doubles from low to high wages — which reads like state-dependence. It is an artefact of *cost curvature*: with a convex effort cost, the same multiplicative β moves more tasks at higher wages. On the *scale-free* log-wedge the wage interaction vanishes, and in the nonparametric bounds the low- and high-stakes intervals overlap. The rule that follows: never read β off task counts — recover it via the two-step, the wage-matching bounds, or the ME-robust set, all of which are curvature-immune. This single artefact would have manufactured the chapter’s flagship “state-dependence” finding out of nothing; catching it is what the Monte-Carlo-first discipline is for.

Lesson 2: honesty about which margins the data can test

The state margins are not all equally testable in-hand. Stakes: cleanly testable (randomized wage grid) — and the verdict is layered: the dual-self/fixed-cost prediction (wedge share shrinking at high stakes) is rejected, and the finer per-quintile test then also rejects *exact* flatness of the scale-free β , in favour of a modest rise toward 1 (Section 2.14.3). Proximity: testable in coarse bins (4–7, 8–14, 15+ days), where a significant gradient *does* appear (Section 2.14) — with the stated caveat that A&R’s design leaves the fine near-region (1–3 days) sparse and calendar/learning effects partially confounded. Load: testable, and tested — measured-sleep bins and randomized cognitive load return *bounded nulls* (Section 2.14.3), framed against the dual-self prediction that load should bite. Cue: not cleanly testable at high dose in any open dataset — the honest remaining gap the micro-survey is designed to seed, with the matched-state requirement of Section 2.14 baked in. A paper that reports where its flagship conjecture is *not* supported (stakes) is more credible on the margins where it is (proximity, visceral states).

2.8 What each dataset delivers (and what is still missing)

Test component	Dataset(s), all in hand and verified loadable	What it delivers
Wedge over effort; naïveté; projection	Augenblick–Rabin 2019; ANS 2015	multi-vantage wedge; predictions $\Rightarrow$ sophistication and projection; wage grid for the RP conditions and the stakes test (the pilot).
Risk switch	A&S 2012b; Ganguly–Tasoff	randomised certain/risky $\Rightarrow$ strips the risk channel.
Liquidity switch	Carvalho 2016; Cassidy 2019; Dean–Sautmann 2021	randomised payday/windfalls, income shocks $\Rightarrow$ strips liquidity (causal in Cassidy/Carvalho).
Proximity / visceral state	Augenblick 2018; Giné 2016; Schilbach 2019	near-region proximity dose-response; sobriety-incentive visceral state $\Rightarrow$ trait-vs-state on the margins we <i>can</i> test.
Noise-layer validation	Kurtz–David 2019 (+2024)	target-vs-click motor data $\Rightarrow$ validates the trembling-hand centering.
Observational arm (ECPF)	Adams et al. 2014 cleaned CSVs + Aguiar–Kashaev Julia pipeline	185 singles / 2004 couples $\times$ 4 quarters $\Rightarrow$ the quasi-hyperbolic-with-measurement-error test on singles (§2.8.1).
Aggregation	Adams 2014; Jackson–Yariv 2014 (lab planners)	singles-vs-couples contrast; “planners” who are present-biased with no individual bias.
Inference & comparison	ARP, Cox–Shi, ELVIS code; Beatty–Crawford	the full three-layer stack, all code in hand.
Plans vs. outcomes at scale	Oscarsson 2020 ($n=1,062 \times 12$ months); Carrera 2022 ($n=1,280$ + gym records); RAND HRS 1992–2022 ($n=12,452$); Canvas Network ($n=325k$)	belief calibration and monthly updating; goal/probability wedges against administrative outcomes; the retirement-plan wedge and its shock decomposition; declared intent at MOOC scale.
Randomised mechanisms at scale	Milkman et al. 2021 StepUp ($n=61,293$, 52 arms)	stakes, framing and lapse-bonus Z ’s; persistence decomposition by mechanism; expert-forecast calibration.
Digital addiction (cue-rich)	Allcott–Gentzkow–Song 2022/23 public microdata ($n=1,933$)	incentivized predictions vs app-measured use (horizon-resolved wedges); randomized bonus/limit Z ’s; limit-valuation certifier; habit persistence.
Deadline and payday effort	OULAD ($n=32,593$, 10.6M click-days); Kaur–Kremer–Mullainathan 2015 (worker $\times$ day)	the proximity gradient in the field with CS/ARP inference; stakes elasticity of the deadline spike; dominated-contract demand.
Commitment demand and efficacy	Hyndman–Bisin 2024; Ashraf–Karlan–Yin 2006; DellaVigna–Malmendier 2006; Sadoff et al. 2020	chosen deadlines vs. completion; commitment-savings take-up vs. hyperbolicity; the contract-choice RP bound; advance-vs-immediate food with the state measured at both vantages.
Field effort-wage curve	DellaVigna–Pope 2018 ²³ ($n=9,861$, 18 arms)	the wage curvature that scales every two-step β , plus delayed-payment arms on the same margin.
Time discounting	Halerman 2015; StudentLife	consistency/stationarity/irrationality

How to go beyond — The one genuine gap, and how it is handled

The one distinctive margin still missing a *purpose-built* dataset is a high-dose randomised *cue/load* manipulation crossed with an intertemporal choice — and even that is now tested, not merely flagged. The load channel returns *bounded nulls* at the doses open data reach: binning Bessone’s workers by measured nightly sleep across its whole range (Section 2.14.3) shows no graded gradient in β_{eff} , and Drichoutis–Nayga’s randomized cognitive load leaves the consistency screen intact — both framed against the dual-self prediction that load *should* bite, so the null is informative rather than empty. The cue-rich digital margin is covered by Allcott–Gentzkow–Song’s public microdata ($n = 1,933$): incentivized predictions against app-measured use, randomized screen-time bonus and limit Z ’s, and a limit-valuation certifier (Section 2.14.7). What remains genuinely open is a *high-dose*, cleanly randomised cue/load design on the same margin; the highest-value buildable targets — session-level game or streaming logs where a quasi-random loss or interruption is the cue, and public pledge-plus-timestamp platforms where intention and action are both dated — are catalogued in the companion data notes as the pre-registered next step.

2.8.1 The observational arm: the Spanish expenditure panel

The chapter has a field counterpart on the very dataset the nonparametric literature was built on. Adams et al.’s cleaned four-quarter panel and Aguiar–Kashaev’s complete pipeline are in hand.

The empty cell in the frontier — and the two-sided verdict it delivered

Cross the two frontier papers: Blow–Browning–Crawford test *quasi-hyperbolic* conditions but deterministically, with no measurement error; Aguiar–Kashaev handle measurement error but test only *exponential* and the collective model. The cell “*quasi-hyperbolic conditions under measurement error*” was empty — and it is exactly this chapter’s question. Filling it (Section 2.14.7): reproduce the AK singles baseline with an independent implementation, put the quasi-hyperbolic restriction into the moment function, and trace the whole (β, δ) grid. The result is sharper than the anticipated one-sided null: exponential discounting fits the singles’ consumption *exactly* (aligning with AK), but the same machinery also *rejects* $\beta \leq .75$ outright — the survey, done right, rules deep present bias out rather than merely failing to find it, and the couples subsample gives the same set even though deterministic tests reject 89% of couples. The two arms were built to fail in opposite directions if present bias were not real; instead each excludes one side, and they overlap only at $\beta \in [0.80, 0.95]$.

2.9 A second, independent channel: anticipation

The future may also pay *now*: imagining it can be pleasant (savouring) or painful (dread), so people pay to time *information* in ways a pure discounter cannot (Loewenstein 1987; Caplin–Leahy 2001). This is a separate test, kept separate. Its signature is a *valence flip*: pay to *delay* good news and to *resolve* bad news early, and choose information-timing that a discounter would be indifferent to. Data are in hand: Ganguly–Tasoff (2016) already show the premise (people pay to open a high-reward envelope early and to avoid an aversive medical test), and Falk–Zimmermann (2024) show the demand is *attention-mediated* — later information becomes attractive precisely when the environment lets one *not attend* to a bad prospect, which is the arbitrator’s attention weight in the model of §2.10. The same moment-inequality machine tests the valence flip. Test B is a complement to the trait-vs-state test, not a substitute; it is developed only as far as existing data support.

2.10 A model of the person — as scaffolding, not a claim

The decomposition and the trait-vs-state test stand on their own; a reader who distrusts grand models can take the empirical chapter and stop. But the state-dependence hypothesis has to come from *somewhere*, and it is worth being explicit about the model of behaviour that generates it — provided we are equally explicit about its status.

What the model is, and what it is not

The *Multi-System Control* (MSC) sketch below is a **framework**, in Browning–Crossley’s sense — like “the life-cycle model,” it is a way of organising particular, falsifiable models, not itself a falsifiable claim. That is not a weakness to disguise; it is a category. The paper does *not* test “MSC.” It tests *restrictions* the framework generates — flat vs sloped β_{eff} , domain/cue specificity, the load prediction, the separation of value-computation noise from bias — each of which is falsifiable and several of which the pilot already puts to the data. MSC earns its place by (i) showing the standard models are special cases rather than rival religions, and (ii) generating the state-dependence prediction that is the chapter’s new content. If the reader prefers, read this section as “why we expect β to be a reaction function,” nothing more.

2.10.1 The architecture

A person is a body served by several *valuation systems* — fast/habitual, visceral/affective, and a slow *deliberative* system (call it D) that evaluates options model-based, with a broad, goal-based horizon. At each moment these systems produce value signals; a fast *arbitrator* combines them with state-dependent weights $\omega_i(X_t)$ and issues an action (a softmax / drift-diffusion blend, in the spirit of Fudenberg–Strack–Strzalecki and Webb 2019). This is the automatic loop.

Answering the question: is the deliberative system the same as the “executive”? No.

Two things that Kahneman’s “System 2” runs together must be pulled apart — and the pilot-grade realism of the model depends on it. The *deliberative system* D is a *valuation* process: it *computes* a careful, future-regarding value and feeds that signal into the arbitrator alongside the others. The *executive* is a *control* process: it does not itself value anything; it decides *whether to engage* costly deliberation and willpower — whether to let D ’s slow valuation run and up-weight it against a clamouring visceral signal. So D is “what the careful part of you thinks”; the executive is “the decision to think carefully and act on it.” Kahneman’s System 2 $\approx D$ *plus* the executive; MSC separates them because, following the expected-value-of-control literature (Shenhav–Botvinick–Cohen 2013) and the model-based/model-free arbitration literature (Daw–Niv–Dayan 2005), *deliberating* and *deciding-to-deliberate* dissociate — you can have an accurate deliberative valuation available and still fail to deploy it. Your reading of the earlier draft was right to flag this; the two are distinct, and the model now says so.

2.10.2 The executive engages only when it is worth it — and a monitor decides when

The executive is costly (willpower is a scarce resource), so it intervenes only when the expected value of control exceeds its marginal cost. But that raises exactly the question a careful reader asks: *who computes whether it is worth it, and is that computation always running?*

The monitor: a cheap, always-on-but-fallible trigger — an enhancement the literature supports

There is a distinct, low-cost *monitoring* process (in the brain, dorsal anterior cingulate / conflict-monitoring; Botvinick, Shenhav) whose job is to *detect* conflict or need-for-control and, when it fires, to recruit the effortful executive. Three properties, each realistic and consequential:

It is cheap but not free, and can be *attenuated* by state.

Sleep deprivation, intoxication, cognitive load, visceral heat, and stress all *lower the monitor's gain* — so it fails to fire even when engaging would be worth it. This is why the drunk, the exhausted, and the cognitively loaded act impulsively *without deciding to*: the executive was never called, because the trigger was dampened (Ward–Mann 2000: load disinhibits; Figner et al. 2010: disrupting lateral-PFC raises impatience).

It can be *wrong*.

The monitor can mis-estimate the value or cost of control — a genuine *error* (mis-calibrated detection), distinct from a mere difference in weights. So “self-control failure” has two routes that a single-parameter β cannot separate: the monitor did not fire (no control attempt) versus it fired but willpower was insufficient (control attempted and lost).

It is sometimes effectively off.

Under strong enough attenuation the executive does not intervene *even when* its expected value exceeds cost — not because the trade-off flipped, but because nothing checked. This predicts that manipulating the monitor's state (load, sobriety, sleep, cue-salience) moves behaviour *holding true preferences fixed* — precisely the state-dependence of β_{eff} the chapter tests, and a second, sharper reason present bias should look like a *state*.

2.10.3 Why the impulse exists, and why it misfires today — the evolutionary microfoundation

The architecture above raises a prior question the chapter should answer rather than dodge: *why is there a hot, cue-driven system at all*, and why does the careful system so reliably lose to it? The oldest human self-descriptions already name the conflict — Plato's thirsty man who is “unwilling to drink,” the serpent's fruit in Genesis (the “animal factor” that, in Tao–Chen's 2023 reading, survives inside us after natural selection), Odysseus lashed to the mast. Naming, though, is not explaining. Evolutionary biology supplies the missing *why*, and it sharpens three of the chapter's empirical claims rather than decorating them.

Intuition — The impulse is not a bug: an ancient controller, well-tuned to a vanished world

In the ancestral environment the cue-triggered approach system was *adaptive*: a ripe-fruit, mating, or novelty cue that fired a fast “take it now” impulse converted directly into fitness, and impatience itself paid, because waiting carried interruption and collection risk (Hayden 2018). Pleasure and payoff were aligned — “live for today” *was* “plan for tomorrow” (Burnham–Phelan 2020). Present bias, in this reading, is neither noise nor a defect of a too-small brain; it is a *routinised, once-optimal decision strategy* (Hayden's “bounded optimality”) running in an environment it was not built for. That is why self-control failures are *systematic* — they go one way, toward the cue — rather than the mean-zero scatter a pure-noise account would give, and it is the substantive reason the chapter models β as a *reaction function* to cues and states, not a coin flip.

Mismatch: modern superstimuli hijack the ancient circuit — and firms engineer them for revenue

The environment then changed far faster than the genes. The cues that trigger the approach system are now *manufactured*: the Big Mac (1967), the slot machine, the infinitely scrolling feed (the iPhone is from 2007) are evolutionarily novel *superstimuli* that light up circuitry tuned for calorie-dense food, for risk-thrill, for social novelty — and now deliver harm where the ancestral versions delivered fitness (Burnham–Phelan’s “mismatch”; Figure 2.5). Crucially the mismatch is *exploited*: an attention, food, and gambling industry reverse-engineers the approach system and designs cues to maximise cue-triggered engagement for revenue — the biological mechanism underneath DellaVigna–Malmendier’s “firms exploit consumer biases.” This is why the cue-saturated digital domain is where the measured wedge is largest and most policy-live, and why the state-dependence of β_{eff} is not an oddity to be assumed away but the *expected* signature of an ancient circuit meeting an engineered cue.

Why money is the cold exception — and why the domain asymmetry follows

The same lens explains the chapter’s sharpest within-person result (Section 2.14.5): *money is evolutionarily recent*. Coins are some 2,600 years old — roughly one hundred generations, far too few for a dedicated approach circuit to have evolved — so abstract currency has no hot immediate component of its own; the visceral pull is on what money *buys* (food, status, mates, thrills), not on the tokens. Effort, by contrast, is a viscerally felt cost *now*. So the fact that the same 101 people carry $\beta_{\text{effort}} \approx 0.88$ but $\beta_{\text{money}} \approx 0.99$ is not a puzzle to explain away; it is precisely what the mismatch account predicts, and it is why a single (β, δ) cannot serve both domains. Boysen’s chimpanzees make the mechanism vivid: shown *symbols* for candy they choose the reward-maximising smaller pile, but shown the *actual* candy they grab the larger one and lose — the approach system fires on the proximal sensory cue, not on the abstract representation. Money is the ultimate abstract symbol; effort and the scrolling feed are the real candy.

Where MSC sits relative to the dual-self model

Fudenberg–Levine’s (2006) dual-self model is the closest formal relative: a patient long-run self pays a costly-willpower “tax” to override a sequence of myopic short-run selves, unifying quasi-hyperbolic time inconsistency, a preference for commitment, *and* Rabin’s small-versus-large risk paradox (Chapter 3) in one object — itself a hint that the impulse/control machinery is domain-general, exploited over time by the attention economy and over risk by the gambling industry. MSC keeps that skeleton and adds three things the empirical chapter needs. (i) The controller *engages selectively*: a cheap, fallible monitor decides *whether* to deploy willpower, so behaviour can move with the monitor’s state (load, sleep, cue) at fixed preferences — the $\beta_{\text{eff}}(X)$ state-dependence. (ii) The distortion is *domain-specific*, attached to the hot immediate component (an approach reward for temptation goods, an avoidance cost for effort), which a single scalar willpower cost cannot generate. (iii) The mismatch microfoundation above says *which* cues fire the short-run self, and why. *Can the data tell MSC apart from the bare dual-self, or is it only a nicer story?* On one margin they cannot be separated, and the chapter says so: *both* predict that depleting the executive worsens self-control, so the bounded null on load (Section 2.14.3) is evidence for neither — the instinct that the load prediction “seems reasonable under MSC too” is exactly right. But two other margins *do* separate them, and one is already decided. **Domain-specificity (decided)**. A single willpower cost implies *one* β per person — the myopic short-run self wants the immediate payoff whatever the domain, so the bare dual-self predicts money

present bias as much as effort present bias. The data reject this within person: $\beta_{\text{money}} \approx 0.99$ against $\beta_{\text{effort}} \approx 0.88$ in the same 101 people ($p < .001$, Section 2.14.5). The dual-self can survive only by *importing* a good-specific hot/cold distinction (money is not “consumed” now, effort-relief is) — but that distinction *is* MSC’s content, delivered by the mismatch microfoundation, not something the bare model predicts. So the within-person money-vs-effort gap is not merely a rejection of (β, δ) ; it is a rejection of the single-willpower-cost structure, and a robust one. **Cue at fixed resources (the decisive, still-open test).** The dual-self’s only lever on self-control is the willpower *resource*; a cue that raises *salience* without depleting resources should therefore do nothing. MSC says a salient cue recruits — or fails to recruit — the monitor, and so moves β_{eff} at fixed resources. A design that *randomises a cue within person* thus discriminates them cleanly: a null favours the dual-self, an effect favours MSC. This is why the *cue* margin, not the load margin, is the frontier worth the data cost — and a within-person micro-randomised cue (a notification, a suggestion, a moment-of-temptation prompt) is exactly the design that settles it.

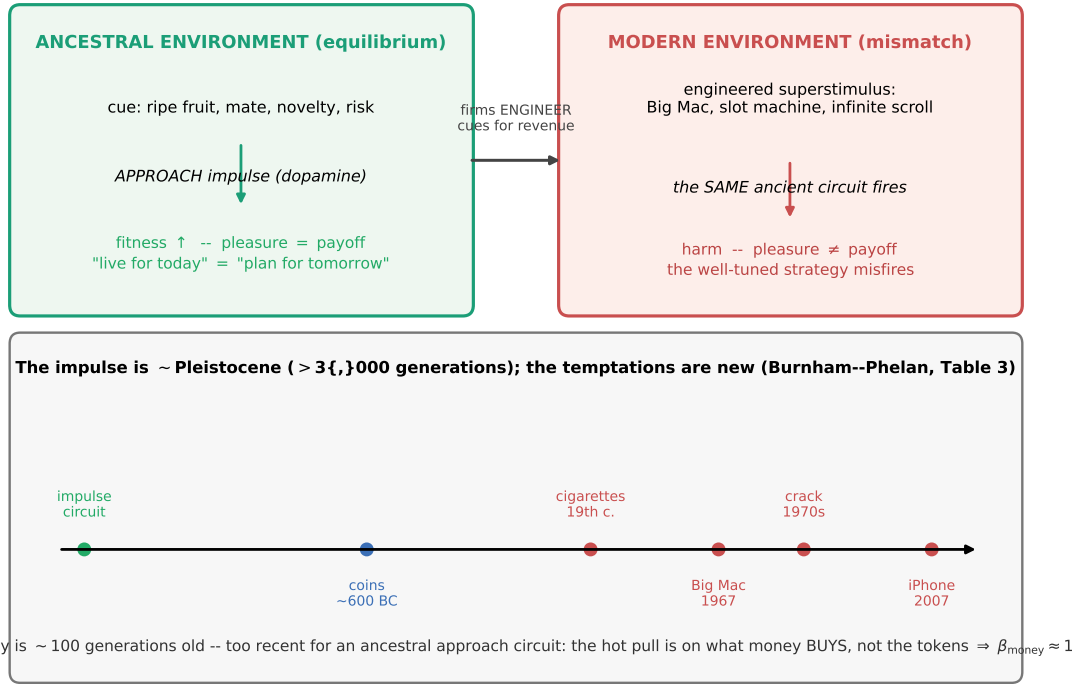


Figure 2.5: The evolutionary-mismatch microfoundation. The same ancient approach circuit that converted cues into fitness in the ancestral environment now fires on engineered superstimuli — deliberately designed for revenue — and delivers harm (Burnham–Phelan 2020; Hayden 2018). The impulse is a once-optimal, routinised strategy misfiring, not noise; money, being evolutionarily recent (~ 100 generations), never acquired a hot component, so the bias lives on what money buys, not the tokens — which is why $\beta_{\text{money}} \approx 1$ while $\beta_{\text{effort}} < 1$.

2.10.4 Present bias is emergent, and the standard models are projections

In MSC, present bias is not a primitive parameter; it is what you *see* when a cue or a hot state spikes the weight $\omega_i(X_t)$ on a myopic system and the monitor fails to recruit the executive in time. A constant β (quasi-hyperbolic) is the reduced form you get if that spike is the same in every state; a *reaction function* $\beta_{\text{eff}}(X)$ is what you get if it is not — which is why “trait vs

state” is the test that discriminates the standard model from this one. Exponential discounting, quasi-hyperbolic, temptation, dual-self, cue-triggered addiction, fixed-cost present bias, and myopia are all *parameter settings or projections* of the same architecture (an arbitrator with fixed vs moving weights; an executive with high vs low cost; a monitor with high vs low gain), not competing faiths — the unifying payoff, and the reason the horse race of §2.4 is a race within one family, not across incommensurable ones.

2.11 Which “you” does welfare serve? — and the honest limits

If the paper is to say a self-control problem is a *problem*, it must say by whose lights. This is where honesty matters most, because the naive move — “the reflective self is right, the impulse is wrong” — smuggles in a value judgement dressed as a fact.

There are no neutral weights — so we do not pretend there are

Any welfare criterion privileges a standpoint, and there is *no fact of the matter* about which is correct. Candidates all have defects: *decision utility* (what you chose) is the very thing in question; *experienced utility* (Kahneman) is hard to measure and, awkwardly, the scrolling *felt good* while it happened; *reflective/endorsed utility* (Frankfurt’s second-order self) is the natural choice but is *post hoc* and possibly contaminated by guilt, hindsight, and an inability to re-inhabit the state one was in; the *long-run self* tyrannises the present. Weights could differ at every moment, and no weighting is objectively “the person.” So the chapter does *not* assert a full welfare function. It makes the weaker, more defensible move.

The Bernheim–Rangel move: claim a self-control gap only where all reasonable standards agree

Rather than resolve the philosophy, identify when *every* plausible criterion points the same way — Bernheim–Rangel’s “unambiguous choice.” Concretely, a self-control gap Γ is claimed for a behaviour *only* when several independent signals converge: the plan-vs-realised wedge is negative, the person *predicted* she would not deviate (naïveté, from the elicited beliefs), she *revealed* demand for commitment *ex ante*, and an informed, unhurried, cool re-elicitation agrees. Where these *converge*, you have a robust case that does not depend on which self is “really” her. Where they *diverge* — she scrolls, enjoys it, buys no commitment, and only feels a vague retrospective guilt — the honest verdict is *agnosticism*: that is **not** a robustly identified self-control problem, and the paper declines to call it one.

Intuition — Why Γ is not the ascetic’s verdict: four answers to the “cold-state bias” objection

The sharpest objection to any plan-based welfare object runs: *the advance plan is just the preference of a cold, bloodless self who underweights pleasures it cannot currently feel — privileging it is asceticism by another name*. The objection deserves a direct answer, and Γ has four. **(1) Γ imports no outside standard.** It compares the same person, with the same information, at two of her own vantage points — not her behaviour against a doctor’s, a moralist’s, or an average. If she plans twelve visits *because she wants to*, and delivers five, the disagreement is internal; Γ merely measures it. **(2) The advance vantage is privileged by arithmetic, not by virtue.** The phrase “ β cancels on both sides” is the crux, so here is the arithmetic. Write the trade-off as an effort cost c borne at date T against its reward u enjoyed at date T' , both in the future. Seen from far in advance (today = 0), quasi-hyperbolic value is $\beta\delta^T(-c) + \beta\delta^{T'}u$: the present-bias factor β multiplies *both* dated terms, so it factors out entirely and the ranking is decided by the exponential

trade-off $\delta^T(-c)$ against $\delta^{T'}u$ — the δ -preference that *every one of her selves shares*, the tempted one included. β cancels because at the advance vantage *neither* side is immediate, so there is nothing for it to single out. Now slide the vantage to the day of the effort: the cost becomes immediate and value is $-c + \beta\delta^{T'-T}u$, where β now multiplies *only* the reward. It no longer cancels; it marks the reward down relative to the cost, and the today-self is willing to abandon the plan. So the advance choice is not a colder or more virtuous preference imposed from outside — it is the one reading of her own preferences on which the planning self and the tempted self *agree*, precisely because β is silent there (Figure 2.3). The today-choice adds exactly one ingredient, the weight on immediate experience — the contested quantity — and Γ measures the disagreement that ingredient creates, nothing more. **(3) The cold state can be biased too — so we test it rather than assume it.** If plans are made hungry (or lonely, or flush), projection contaminates the *plan*, and the advance vantage loses its privilege. That is precisely why the design measures the state at *both* vantages where it can (Sadoff’s panel records hunger at the order *and* at the delivery; the Monte-Carlo’s matched-state requirement is the general form) and why lesson (i) qualifies every unmatched gradient. The asceticism worry is, in our language, the projection channel — a testable masquerade, not a philosophical stalemate. **(4) Γ alone never carries a welfare claim.** The Bernheim–Rangel discipline above requires convergence — the wedge, the person’s own incentivized beliefs, and the model-set verdicts must agree — before anything is called a self-control *problem*. Where the immediate pleasure is real and the person, seeing her own numbers, endorses the behaviour, the criteria diverge and the chapter’s verdict is agnosticism. In short: Γ is the person disagreeing with herself, measured; asceticism would be us disagreeing with her, assumed.

Intuition — Why this is more than the obvious “self-control gaps are bad,” and the scrolling case

It might seem trivial that a self-control gap is bad and should be closed. It is not, for two reasons. First, the naive view *assumes* we know the reflective self is right; MSC’s three-utilities distinction plus the Bernheim–Rangel test turn that assumption into a *measurable, refutable* claim — it tells you *when* you may call something a mistake and, crucially, *when you may not*. Second, it takes your own scrolling example seriously: the post-hoc “that wasn’t me” is genuinely suspect — after the fact the craving is gone, so of course you disown it; that feeling alone certifies nothing. The fix is not to trust retrospection but to require *ex-ante* revealed commitment plus convergence. Under that standard, a great deal of what paternalists would “correct” does *not* qualify, and the policy implication is to *do nothing* and respect the choice. That restraint — knowing when the model licenses intervention and when it forbids it — is the welfare contribution, and it is the opposite of obvious.

Why it matters — What the decomposition plus the welfare test buys, concretely

The two questions map to distinct policies, and the framework says which regime a behaviour is in. **Mostly liquidity** (the wedge collapses when the budget is loosened) $\Rightarrow$ credit or transfers, not commitment. **Genuine, sophisticated, trait** present bias with converging $\Gamma \Rightarrow$ offer commitment devices and generic defaults (they will be used). **Genuine but naïve** $\Rightarrow$ information and nudges, because the person will not self-correct or buy the device. **Genuine and state-dependent** (if a cue/load slope is found) $\Rightarrow$ add cue management to the menu — notification defaults, friction, availability rules — *alongside*, not instead of, person-level tools. **Myopia, or criteria that disagree** $\Rightarrow$ improve information/perception, or do nothing. In digital self-control, for instance, measured “addiction” that is mostly myopia with no commitment demand calls for design defaults,

not taxes; genuine sophisticated present bias calls for screen-time commitment tools; naïve present bias calls for informing people about their own future use. Same measured behaviour, different measured composition, different policy — which is the entire point of measuring the composition.

2.12 Application domains

The method is domain-general: the same decomposition and trait-vs-state test apply to drugs, alcohol, betting, diet, saving, exercise, deadlines, and the alarm clock. The broad policy relevance comes from that generality, and the paper does not need a single “headline” domain. One is worth flagging as especially current and data-ready without elevating it above the rest — *phone / social-media use*: high-frequency, cue-saturated, attention-consumed-at-use, and policy-live, with two datasets in hand (Allcott–Gentzkow–Song 2022, whose screen-time-limit treatments are *Z*-like switches and whose willingness to pay for limits is field commitment demand; and Allcott et al. 2020’s Facebook-deactivation welfare experiment). Other high-stakes field settings — sports-betting volumes and legalisation timing, payday-loan and savings-commitment data — scope the same extension. In every domain the questions are identical: how much of the measured impatience survives the switches, is it trait or state, and do naïfs under-demand the commitment tool.

2.13 Threats, contribution, and roadmap

The honest threat list

External validity: the core evidence is lab effort; a clean $\beta_{\text{eff}}(X)$ in the lab need not govern real addiction. Handled by the field arm, explicitly graded quasi-experimental — a two-stage claim. **Power:** at short panels (ECPF, four quarters) revealed-preference tests are weak — and the executed field arm confirms it *quantitatively* (the Monte-Carlo puts deterministic power near zero without price variation) while still delivering a two-sided verdict: the ME-robust set on the ECPF *excludes* deep present bias ($\beta \leq 0.75$ rejected) even though it cannot separate time consistency from moderate bias. The field arm bounds; it does not point-identify — stated as a finding, not an apology. **Observational equivalence:** models are close on any one margin; handled by set-identification and a predictive-success horse race, not point identification (Ericson–Laibson’s map). **The cue/load gap:** the most distinctive state margin needs data no open source supplies; handled by testing the margins we have, seeding a micro-survey (with the matched-state design requirement of Section 2.14), and pre-registering the experiment. **Commitment as a contaminated certifier:** confirmed empirically (correlation $+0.02$ with the wedge it should certify); welfare rests on the plan-vs-realized wedge Γ , never on commitment alone. None is fatal; each is stated rather than hidden.

The contribution, precisely

Three things are genuinely new, each now *executed* with results in Section 2.14. (1) **Trait vs state:** the first framing of present bias as a reaction function $\beta_{\text{eff}}(X)$ with a nested revealed-preference test of its flatness — delivered: *near-flat but not exactly flat on stakes* (the finer per-quintile test rejects exact flatness, $p = .001$, in favour of a modest rise toward 1 — the magnitude-effect direction — while the fixed-cost collapse stays excluded), *sloped on proximity* (GMS $p = .007$ in the lab; deadline and payday gradients at CS/ARP $p < .001$ in the field), *live on visceral and liquidity margins, bounded-null*

on executive margins, and the cue/load cell scoped honestly for the micro-survey. **(2) Inference on the type mix:** set-identified population shares delivered on real subjects — time-consistent [0.67, 0.79], genuinely present-biased naïfs [0.09, 0.32], sophisticated [0, 0.11] — using design-calibrated mixing matrices and exact-size moment-inequality inference (GMS, Cox–Shi, ARP, all Monte-Carlo-certified head-to-head). **(3) The noise-and-aggregation closure:** the quasi-hyperbolic conditions *with measurement error handled* run on the effort panels (the first noise-robust nonparametric present-bias sets over effort, dropping nobody: A&R [0.75, 0.95], Fedyk interior [0.80, 0.95], both excluding time consistency) and on the ECPF singles (the empty cell in the Blow–Browning–Crawford $\times$ Aguiar–Kashaev matrix: [0.80, 1.00], excluding deep present bias), with aggregation treated as a channel and shown to die under noise-robust inference. A fourth element is the β – δ **symmetry audit** (Table 2.1): rather than ask “is $\beta < 1$,” the chapter asks whether *one* β can serve a person’s every choice, and shows it cannot — effort is present-biased where money is not in the same people ($\beta_e = .88$ vs. $\beta_m = .99$, $p < .001$), β drifts with stakes, with proximity, and reverses under anticipation; four of five symmetries rejected, each by the taxonomy’s assigned contrast, and MSC predicts exactly that pattern. A fifth element grew out of executing the rest at field scale: **the welfare object measured, its certifiers broken:** plan-vs-outcome wedges of +10 to +38 percentage points across resolutions, gym goals, contracts and retirement plans; belief updating at a quarter of the Bayesian rate; commitment demand uncorrelated with the wedge it should certify in six independent datasets (and inert on the deadline margin where its efficacy is testable); expert forecasts of behaviour-change miscalibrated eightfold; and persistence after incentives that is mechanism-specific, not a habit stock. The organising scaffold (MSC) and the disciplined welfare stance (Bernheim–Rangel Γ , claimed only on convergence) tie the empirical results to policy without overclaiming.

Order of execution, every step on data already on disk: (0) upgrade the Monte-Carlo to validate size and power of the decomposition, the trait-vs-state test, and the noise layer, with the seven channels and the four off-model rivals as agents; (1) the effort reanalysis (the pilot, extended: full decomposition on ANS/A&R, the scale-free β_{eff} on every available margin); (2) the ECPF arm (reproduce Aguiar–Kashaev, add the quasi-hyperbolic restriction on singles); (3) the confound-switch portfolio (risk, liquidity, visceral) and Test B; (4) the field extension across exercise, learning, paid work, food, and life-cycle plans — plans vs. outcomes, commitment demand vs. efficacy, mechanism-specific persistence (all executed, Sections 2.14.4–2.14.7); (5) the optional load micro-survey; (6) write. A purpose-built cue/load experiment, and a cue-rich digital-exhaust panel, are the natural sequels.

2.14 From plan to evidence: procedure and results

Everything in this section is *executed code*, not a proposal. The pipeline lives in `code/final/`: a shared inference engine (`toolkit.py`), a Monte-Carlo that certifies every test on synthetic populations where the truth is known (`mc_final.py`, blocks T1–T10), the experimental arm (`exp_final.py` + `exp_v2.py` + `exp_v3.py`: the effort panels plus the plans-beliefs-commitment batch — Oscarsson, Carrera, Hyndman–Bisin, Sadoff, DellaVigna–Pope, Halevy), and the field arm (`field_final.py` + `field_v2.py` + `field_v3.py`: the ECPF on Aguiar–Kashaev’s exact input, plus the StepUp megastudy, OULAD, the RAND HRS, Kaur–Kremer–Mullainathan, DellaVigna–Malmendier, Dean–Sautmann, SEED, Canvas, StudentLife, and Allcott–Gentzkow–Song’s digital-addiction experiment) — thirty datasets in all. Two disciplines govern all of it. *First, no test touches data before the Monte-Carlo certifies its size and power on agents designed to fool it. Second, every estimate is reported at three rungs of decreasing assumptions* (Figure 2.6): a scale-free two-step point estimate (isoelastic cost, subject-cluster bootstrap);

nonparametric **bounds** from wage matching (only monotone advance supply — no cost function at all); and a measurement-error-robust **confidence set** (an ELVIS-type existence test whose only behavioural input is the model’s feasibility conditions and whose only statistical input is centered measurement error). When all three rungs agree, the finding does not depend on what we assumed — and, as the figures below show, they agree.

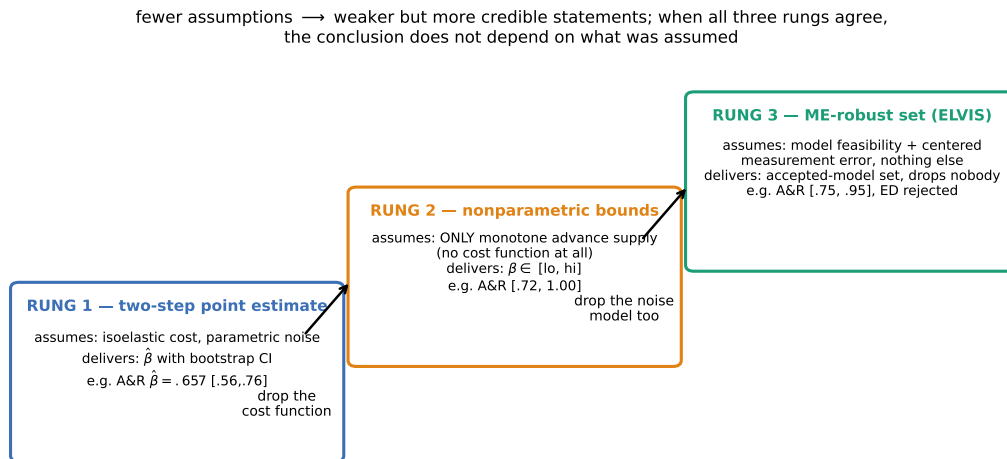


Figure 2.6: The three-rung reporting ladder. Climbing right drops assumptions: the two-step needs a cost function; the matching bounds need only monotonicity; the ELVIS set needs only the model’s feasibility conditions plus centered measurement error — and drops no subject. Every headline number in this section is reported at all rungs that its data permit.

Because this section reanalyses other people’s data, one distinction must stay visible throughout: *what is reproduced* (to anchor and validate against the original paper) versus *what is new here*. Table 2.2 keeps the ledger; in the text, reproduced benchmarks are always attributed (“= the paper’s headline”), and everything else — the conditional CS/ARP inference, the bounds and ME-robust sets, the certifier correlations, the horizon and vantage decompositions, the persistence and calibration layers — is this chapter’s addition.

Table 2.2: The reproduce-vs-beyond ledger for the main datasets.

Dataset	Reproduced (anchor)	New here (ours)
A&R 2019	their MLE $\beta = .83$ (cited)	scale-free two-step; wage-quintile joint CS/ARP tests; stakes/proximity conditional wedges; nonparametric bounds; drops-nobody ME-robust set [.75, .95] on the full sample; per-quintile $\beta(\text{stakes})$ trend test
ANS 2015	pooled wedge sign	decomposition rows; certifier correlation; conditional price-bin tests
Fedyk 2024	naïveté direction	the triangle at two rungs; interior-unit ME-robust set; censoring artefact certified by MC
Cerrone 2026	their MLE range (bracketed)	day-FE absorption diagnosis; type-share LP bounds with error-calibrated mixing
ECPF (AK 2021)	ED accepted for singles ($TS = 0$)	the QH-with-ME cell: full (β, δ) grid, $\beta \leq .75$ rejected; couples contrast under noise
StepUp 2021	arm-level ITTs (sign/scale)	mechanism families; stakes/framing/lapse contrasts; persistence Wald; expert-forecast calibration by family
OULAD	— (descriptive dataset)	deadline-distance gradient with CS/ARP; stakes elasticity of the spike; calendar quasi-experiment; IMD margin
HRS	aggregate calibration (known)	individual wedge; person-clustered joint test; horizon profile; shock decomposition; FT-vs-any margin; revision drift
Carrera 2022	overconfidence direction	calibration gaps with joint tests; certifier correlations; recall-bias channel
KKM 2015	payday effect exists	within-worker profile with CS ordering test; trait-behaviour certifier matrix
DellaVigna–Pope	arm means	effort–wage elasticity as the two-step’s exchange rate; delay DiD; matching bound; the flat-curve instability demo
AGS 2022	bonus/limit ITTs (= their Fig. 4/5)	horizon-resolved prediction wedges with CS/ARP; fixed-target revisions; updating slope; certifier #6 (valuation vs measured wedge); habit-naïveté contrast

2.14.1 The Monte-Carlo certificate

The simulator implements a twelve-configuration agent zoo — exponential, naïve/sophisticated/partially-sophisticated present bias, fixed-cost and temptation agents, risk-confounded, liquidity-confounded, projection-biased, state-dependent β_{eff} , myopic, domain-specific, and aggregated households — always with measurement error and preference heterogeneity on. Table 2.3 summarises; three exhibits deserve prose.

Table 2.3: Monte-Carlo validation (200 reps, $n = 120\text{--}150$, nominal size 5%).

Test	Decisive scenario	Size	Power
T1 decomposition	risk/liquidity agents fool the <i>raw</i> wedge test 100% of the time; the controlled-cell test clears them	.035–.045	1.00
T2 type-share bounds	LP bounds on a 5-type mixture	coverage $\geq .975$	widths .18–.47
T3 trait vs state	state ladder (.90, .75, .60) recovered exactly; Wald flatness at exact size	.055–.07	1.00
T4 stakes contrast	additive/fixed-cost family detected; multiplicative β flat	.055	1.00
T5 sophistication	belief- β recovered at 1.00/.85/.70 for naïve/partial/sophisticated	.045	1.00
T6 domain coherence	$\beta_e = \beta_m$ vs. domain-split agent (delta-method t)	.07–.08	1.00
T7 noise layer	deterministic RP falsely rejects a true exponential 83% under 10% ME	—	—
T8 inference engines	GMS vs. Cox–Shi vs. ARP, head-to-head	.035–.08	1.00
T9 nonparametric bounds	coverage under isoelastic <i>and</i> non-isoelastic cost	1.00	width $\approx .40$
T10 ME-robust sets	truth-in-set .92–1.00; ED discriminated 1.00/0.00; corner subjects included	—	—
T11 quantile vs linear cond.	a middle-of- Z effect: linear slope blind, quantile bins catch it	.04/1.00	—

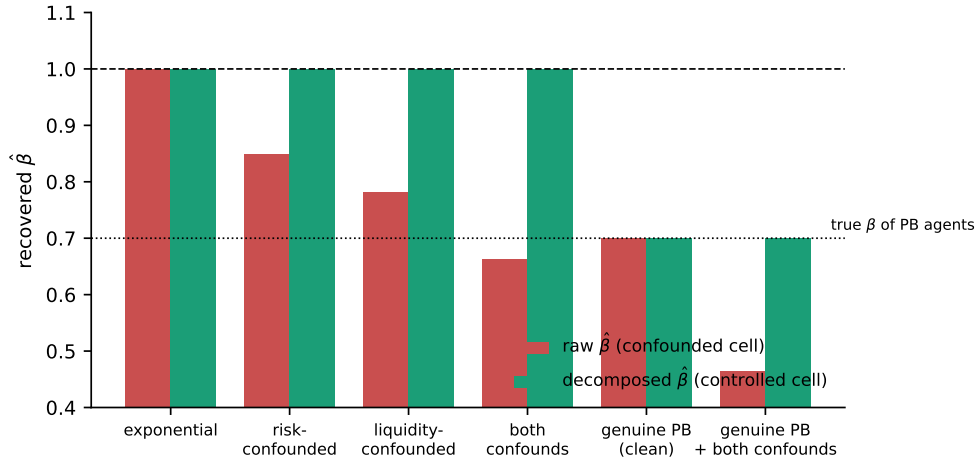


Figure 2.7: The decomposition arithmetic, certified. Each pair of bars is one synthetic agent type: the *raw* wedge estimate (red) reads confounds as present bias — a pure risk- or liquidity-confounded exponential agent measures as $\hat{\beta} \approx 0.66\text{--}0.85$, and a genuinely present-biased agent carrying both confounds measures as 0.46; the controlled-cell estimate (green) returns each agent to its true value (1.0 or 0.7). Conflation roughly doubles apparent bias, and it is fully reversible by design.

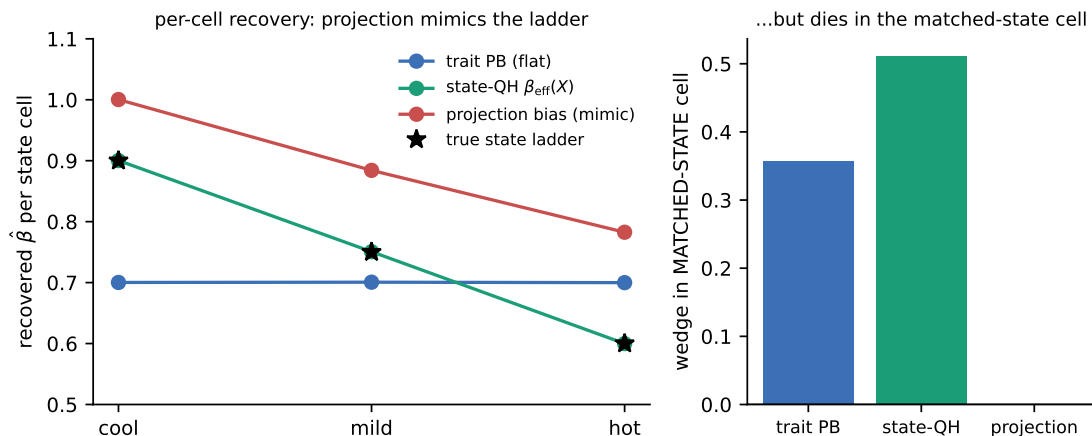


Figure 2.8: The trait-vs-state test and its hardest rival. Left: per-cell $\hat{\beta}$ across cool/mild/hot state cells — the state-QH agent’s ladder (.90, .75, .60) is recovered exactly, the trait agent is flat at .70, and *projection bias mimics the full gradient* with no true present bias. Right: the discriminator — an advance decision made *in the same state* as the workday kills the projection wedge ($G = 0$) but not the genuine ones. Matched-state cells are therefore a hard design requirement for any state-dependence claim.

Three Monte-Carlo findings are results in their own right. **The BBC power warning reproduces in-house:** with one good and no price variation, the deterministic revealed-preference test has *no* power against quasi-hyperbolic truth (rejection .02) while over-rejecting a true exponential at .83 under measurement error — structure, not determinism, buys power (T7); with price variation and the ME-robust machinery, size is exact and power returns (T10). **Aggregation fakes only mild bias:** households mixing exponential members with discount factors as far apart as (.98, .30) measure pseudo- $\beta = .78$; at plausible heterogeneity the artefact lives in the .90–.99 range and in wedge moments, not feasibility sets (T7c). **And the machinery is honest about its own traps:** quadratic-form tests need non-redundant moments (encode ordering nulls by adjacent increments); censored choices break centered-ME moments mechanically — both caught in simulation before touching data, both now coding rules.

Why every conditional test in this chapter bins Z into quantiles, not a slope

A recurring choice in the design layer is how to condition a wedge on an instrument Z (wage, proximity, hunger, deprivation). The tempting default — regress the wedge on Z and read the slope — assumes the effect is *linear* in Z and answers only “does the average wedge move with Z .” The chapter instead splits Z into quantile bins and runs the moment-inequality machinery (Cox–Shi/ARP) on the vector of bin means. This buys two things a slope cannot: the stronger, more credible statement “the wedge is present in *every* bin” (a conjunction of inequalities, not an average), and power against *non-monotone* state-dependence. Block T11 makes the cost of the linear shortcut concrete (Figure 2.9): when present bias appears only over a *middle* range of Z — exactly the MSC prediction that $\beta_{\text{eff}} \rightarrow 1$ at both very low stakes (not worth attending) and very high stakes (the controller engages) — the linear slope test rejects only 4% of the time (blind), while the quantile test rejects 100%; both hold their 5% size under a flat null, and when the effect really is monotone the linear test is only trivially more powerful (1.00 vs. 0.985). So the quantile-binned conditional test is the primary object throughout; the linear slope is reported only as a one-number summary. This is the Andrews–Shi conditioning philosophy, and it is why the stakes verdict

of Section 2.14.3 is a per-quintile profile rather than a single interaction coefficient — the coefficient, we now know, could have hidden a U -shape.

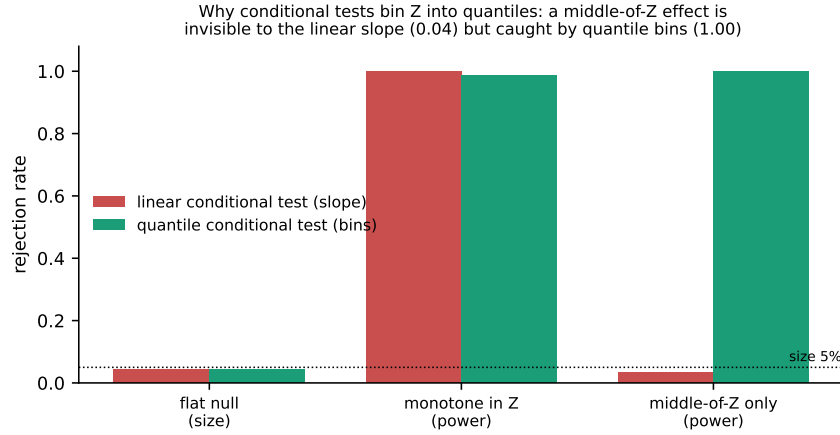


Figure 2.9: Quantile-binned vs. linear conditional tests (Monte-Carlo T11, 400 reps). Under a flat null both control size; when the wedge is monotone in Z both have power; but when present bias is confined to a *middle* range of Z , the linear slope is powerless (.04) while the quantile test has full power (1.00). Conditioning on Z nonparametrically is the primary tool; the linear slope is a summary that can hide non-monotone state-dependence.

2.14.2 How much of the wedge is genuine — and how large is β

Start with existence and provenance. On A&R's effort panel the plan-vs-today wedge is positive *in every wage quintile jointly* (Cox-Shi $p = .004$, GMS $p = .001$, ARP $p = .072$) and *flat across quintiles* (Wald $p = .72$): present bias on every price slice, price-driven on none — the conditional-on- Z version of the genuine-PB test that the raw literature never runs. On ANS the same conditional test is honest about power (positive in every price bin, jointly $p = .23$ at $n = 80$); the pooled wedge stands at -1.30 tasks ($t = -2.0$, 61% of subjects negative). The risk and liquidity channels are confirmed as *real* masquerades with in-hand experiments: certainty contrasts of $+50/-15.4$ tokens ($t = 13.3/-6.3$) in Andreoni-Sprenger's risk arms, and cash-on-hand raising log-productivity by .048 (se .012) in Kaur et al.

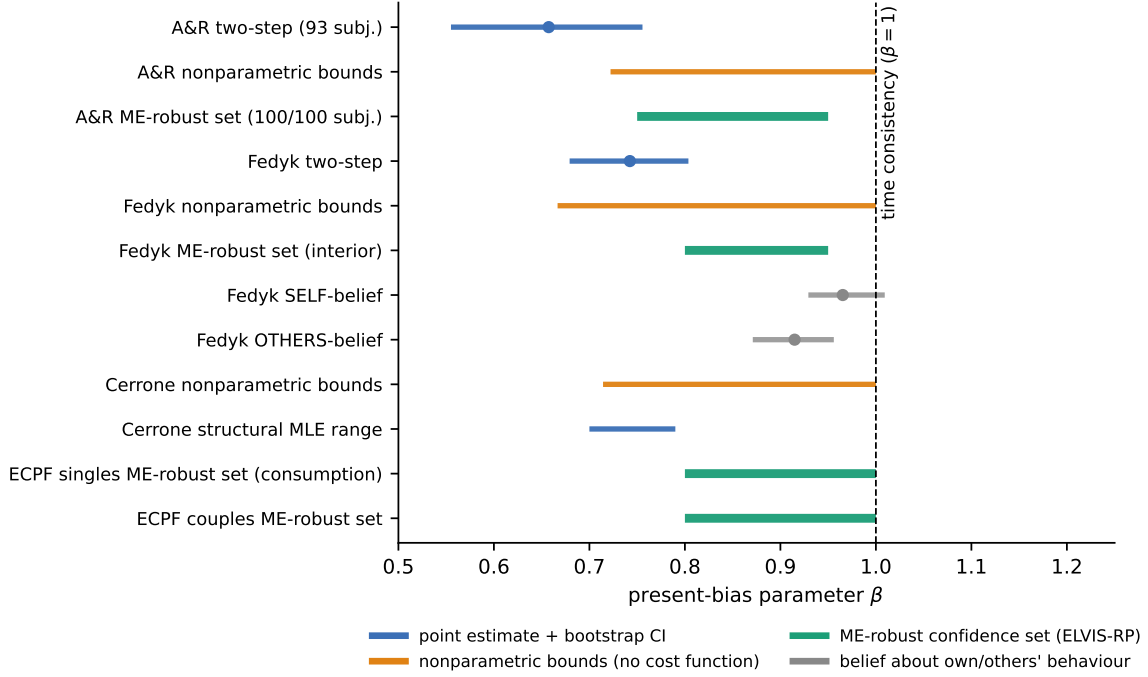


Figure 2.10: The β landscape: every estimate at its three assumption rungs. Blue: scale-free two-step point estimates with subject-cluster bootstrap CIs. Orange: nonparametric wage-matching bounds (no cost function; conservative CI-widened brackets). Green: measurement-error-robust confidence sets (ELVIS-RP; latent choices from the model’s feasible set, centered-ME moments only; *no subject dropped*). Grey: beliefs. The effort sets exclude time consistency; the consumption set (ECPF) excludes deep present bias; they overlap only at $\beta \in [0.80, 0.95]$.

The headline magnitudes, at all three rungs (Figure 2.10). **A&R**: two-step $\hat{\beta} = .657$ [.555, .756]; bounds [.72, 1.00] (93 subjects, all kept); ME-robust set **[0.75, 0.95]** on the *full* 100-subject sample — where A&R’s structural estimation drops 28% — with exponential discounting rejected ($TS = 33$ vs. crit 18.3) and A&R’s own MLE (.83) inside. **Fedyk**: two-step .742 [.679, .804]; bounds [.67, 1.00]; ME-robust set **[0.80, 0.95]** on interior subject $\times$ task units — a second, independent noise-robust effort set nearly coinciding with A&R’s. (On *all* units the set is empty: Fedyk’s rounds are censored at 0, and the Monte-Carlo’s censoring lesson says exactly this must happen — the empty set is a certified artefact, reported as such.) **Cerrone**: bounds [.71, 1.00] bracketing their structural MLE range [.70, .79] from a five-point wage grid; our two-step .78. Four independent designs, one method, one answer: *over effort, β lives in the high-0.7s to low-0.9s, and time consistency is excluded without any functional form and without dropping anyone.*

2.14.3 Trait or state: the reaction function on every testable margin

Stakes: the question decomposed — and a verdict our own sharper test revised. “Stakes” hides two distinct objects, and the tests must be sorted by which one they move. (a) *The cost magnitude* — how much effort is at stake. Here the discriminator is the *shape* of the wedge (T4): a fixed-cost/temptation agent’s absolute wedge is constant, so its share collapses as the stake grows; a multiplicative β scales. The data reject the collapse: across A&R’s wage quintiles the absolute wedge *grows* from 2.3 to 8.8 tasks while the log-wedge stays in a narrow band (.04–.12) — the per-unit distortion is roughly stable, the total damage scales with what is at stake. A flat β never meant “stakes don’t matter”; it means the damage is *proportional*. (b) *The reward magnitude* — is β itself a function of the wage, as the magnitude-effect literature

(Thaler 1981; Noor 2011’s magnitude-dependent discounting) suggests? Our first pass used the single linear interaction (+.007, se .029) and read flat. The sharper instrument built for this revision — per-quintile scale-free $\hat{\beta}$ ’s with a common curvature and joint subject-bootstrap inference (Figure 2.11) — **rejects exact flatness** (Wald $\chi^2_4 = 17.8$, $p = .001$) and finds a **modest positive trend**: $d\beta/d\ln w = +.072$ [+ .011, +.144], one-sided $p = .016$ — β drifts *toward* 1 as the piece rate rises: the magnitude-effect direction, on the effort margin, net of curvature. This *reconciles* rather than contradicts the money literature: the classic money magnitude effect largely dissolves once utility curvature is corrected (Andersen et al.), and Sun–Potters (2022) find CTB allocations moving with budget size but *no present-bias non-stationarity at any magnitude* — a curvature/background-wealth phenomenon that the scale-free design nets out, leaving exactly the small residual trend we measure. The field results are complementary but are *not* direct β (stakes) tests, and are now labelled accordingly: DellaVigna–Pope’s tenfold piece-rate arms trace the effort–wage *curve* (the two-step’s denominator: +7.3% per 10 $\times$); OULAD’s grade-weight elasticity of the deadline spike (.084, $z = 5.1$) is small and positive — coherent with the modest-trend reading; StepUp’s higher-incentive arms move an *immediate* reward (+0.6pp on a +1.6pp base), informative about reward salience, not about the intertemporal stake. *Why the response is small, and where it should become large*: in the MSC reading the controller engages when the stakes are salient enough to be worth the override — nobody procrastinates their own wedding — so β_{eff} should approach 1 as stakes grow, and the measured +.07 per log-wage is the small-stakes tail of that prediction; the tested range (1–35 cents per task, grade weights, cent-scale bonuses) cannot reach the asymptote, so the claim is bounded and stated as such. A relative-stakes probe (wage against the subject’s own effort capacity) finds the trend no steeper for low-capacity subjects, so within this range the drift is not stakes-relative-to-scale; a stakes $\times$ income field test is flagged for the high-stakes sequel. The nonparametric halves-bounds ([.73, 1.00] low vs. [.72, .88] high) remain consistent with a modest trend and exclude a dramatic one.

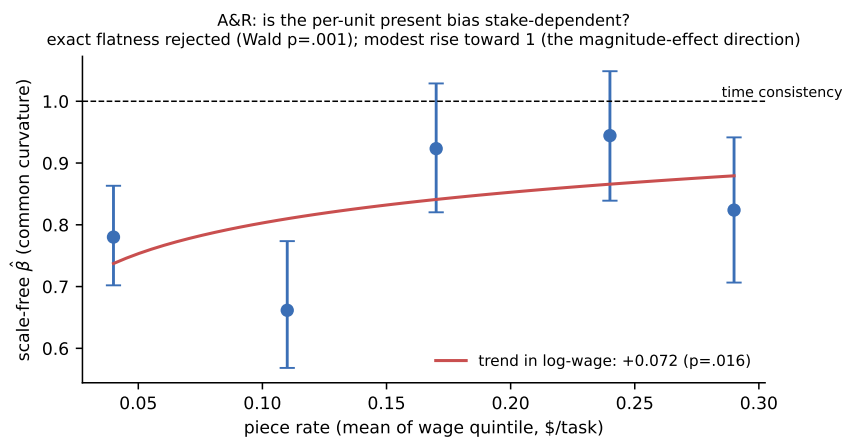


Figure 2.11: The stakes question, done properly: per-wage-quintile scale-free $\hat{\beta}$ on A&R with joint bootstrap CIs. Exact flatness is rejected ($p = .001$); the trend is a modest rise toward time consistency at higher stakes — the magnitude-effect direction — while the fixed-cost prediction (a wedge share collapsing in stakes) remains excluded.

Proximity: sloped — in the lab and in the field (Figure 2.12). Among A&R’s future decisions, jobs fall as the workday nears (−.205 at 4–7 days, −.116 at 8–14 days, relative to 15+; GMS ordering $p = .007$), and the gradient survives in bounds form. The field twins are new to this literature’s toolkit: OULAD students’ daily study clicks *double* as an assessment deadline approaches (10.7 vs. 5.4 clicks/day at 0–2 vs. 14–27 days out; flatness rejected, Cox–Shi and ARP $p < 10^{-6}$, 28,516 students; 29.5% of submissions are late and 33% arrive within 24 hours

of the deadline), and Kaur–Kremer–Mullainathan’s data-entry workers produce measurably more as payday nears (flatness rejected, Cox–Shi $p = .0005$; the shape is noisy and cash-on-hand also cycles with payday — stated as such). The proximity margin, unlike the stakes margin, is *alive* everywhere effort is dated.

Visceral and liquidity states: live — with both vantages now measured once. Hunger shifts within-person discounting ($\Delta \log k$: food +3.03, $t = 6.8$; money +1.14; music +.89 — Skrynka–Vincent); cash-on-hand moves productivity (+4.8%, Kaur et al.); money-domain present bias appears in a low-income population (all three domains positive, 697 subjects, Cheung et al.). Two field additions discipline the reading. In Sadoff–Samek–Sprenger’s grocery panel the state is recorded at *both* vantages — hunger when the advance order was placed and hunger when the immediate choice was made — which is exactly the projection-vs-state-QH separation the Monte-Carlo demands; at $n = 177$ both loadings are bounded nulls (−13 and +13 kcal, $se \approx 13$ –18), so the design point stands even where this sample cannot decide it. And in Dean–Sautmann’s Mali panel the authors’ own present-bias flag is as often future-biased as present-biased (13.1% vs. 14.7%) and moves with the savings stock in the liquidity direction (−.024 [−.066, +.013]) — the Z_{liq} channel live in a real consumption panel, if bounded.

Executive resources: the prediction the leading model makes, tested — a bounded null. The dual-self model (Fudenberg–Levine 2006) predicts that *depleting* the executive should lower β — Shiv–Fedorikhin’s subjects, made to hold a seven-digit number, switch from the fruit salad to the chocolate cake. The chapter tests that prediction rather than assuming it away, and tests it with a *Z-binned* conditional design, not a single treatment dummy that a threshold could slip past. Binning Bessone et al.’s 425 Chennai workers by objectively measured nightly sleep across its whole range (3.4–8.0 h), β_{eff} is non-monotone and essentially flat (.89/.88/.85/.95 by quartile; slope $d\beta/d(\text{sleep hr}) = +.016$, $t = 0.9$; baseline-differenced +.017, $t = 0.7$; joint monotone-departure $p = .36$) — no graded depletion gradient anywhere in the distribution. The randomized sleep and nap arms move it no more ($\Delta\beta = +.002$ /.058), and randomized cognitive load leaves the *consistency screen* intact (CCEI .903 vs. .894, $t = .8$; GARP violations equal — Drichoutis–Nayga). So at the doses these studies reach, the executive-resource channel is a *bounded null*: the Fudenberg–Levine depletion prediction is not borne out, and the *Z-binned* test makes the bound sharper than an average could — it would have caught a threshold and finds none. The honest caveat is “these doses”; a high-dose randomized-load×intertemporal design remains the clean frontier test. A related null from the megastudy below: a *randomized bonus for returning after a missed workout* fails to lift the post-lapse return rate (−0.3pp [−2.9, +2.3] against the same incentive core), and loss- vs. gain-framed micro-incentives of equal expected value tie (+0.4pp [−2.9, +3.7]) — the lapse trap is not a small price problem, and the wedge is not a framing artefact.

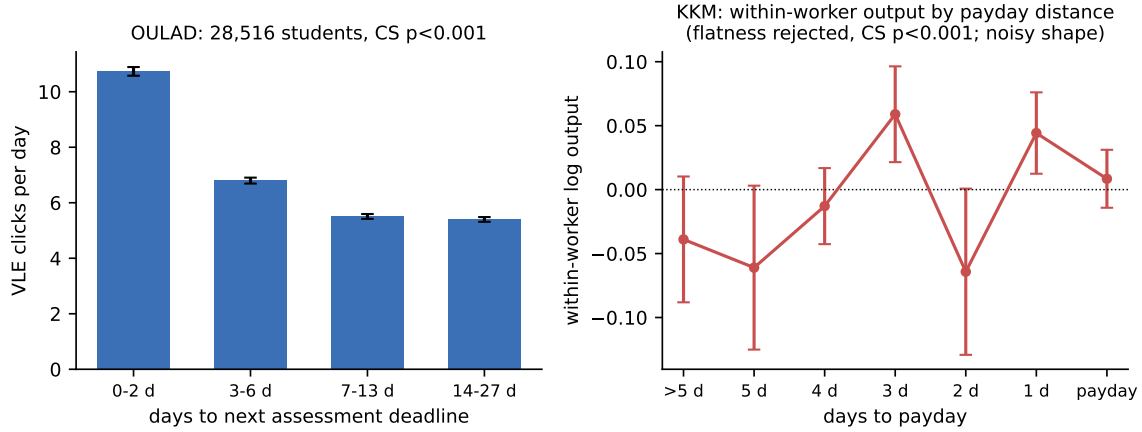


Figure 2.12: The proximity gradient in the field. Left: daily virtual-learning-environment clicks by distance to the next assessment deadline (OULAD; 28,516 students; flatness rejected, CS/ARP $p < 10^{-6}$; the near/far ratio is 1.99, and its elasticity to the assessment’s grade weight is only .084). Right: within-worker log daily output by distance to payday (Kaur–Kremer–Mullainathan; flatness rejected, CS $p = .0005$; noisy shape; cash-on-hand cycles with payday, so this is $\beta_{\text{eff}}(\text{proximity})$ *cum* liquidity).

The pattern is the MSC arbitration picture of Section 2.10: states move behaviour through *value weighting* (visceral, liquidity, population), not through broken *consistency* (load leaves CCEI intact) — and the process signatures agree at database scale: confidence rises with subjective-value distance (+.041, se .006; Bulley et al.) and response time falls with distance to indifference (−.184, se .008; 1.01M trials, 9,605 subjects, 100 papers). Intertemporal choice behaves like graded value comparison, not a binary override. The same graded picture shows up in the StudentLife sensing panel: when deadlines close in, total phone engagement *rises* (+.08 log points, student fixed effects — the phone is also a work tool) while entertainment and social-app use *falls* (−.07) — temptation yields to pressure by displacement, not by an on/off override. The one margin the Monte-Carlo says we *cannot* yet claim: any state gradient estimated from unmatched data (including proximity above) has a projection-bias reading; only matched-state advance decisions separate them (Figure 2.8), which binds the micro-survey design — and which the Sadoff two-vantage measurement shows is *implementable* in a field panel.

2.14.4 Beliefs, plans, and the welfare object: the wedge is everywhere, and nothing certifies it

Fedyk’s data, our estimator, three objects the literature conflates — now separated at two rungs each: actual behaviour ($\hat{\beta} = .742$ [.679, .804]; bounds [.67, 1.00]), beliefs about *self* (.965 [.929, 1.009] — the CI includes 1 and is disjoint from the actual- β CI: naïveté), and beliefs about *others* (.915 [.871, .956] — excludes 1: sophistication about everyone but oneself). People can see present bias fine; just not in the mirror.

Two readings of that asymmetry deserve to be on the record. *Naïveté or motivated optimism?* The optimistic self-belief could be a sincere forecasting error, or a functional asset — self-confidence maintained *because* it motivates (Bénabou–Tirole; Brunnermeier–Parker’s optimal expectations). Incentivized elicitation does not separate them: in both Fedyk’s design and the AGS accuracy-reward randomization, paying for accuracy leaves the bias intact, which rules out cheap talk but not self-deception that survives stakes. For the welfare wedge Γ the distinction is immaterial — both are wrong beliefs held against evidence, and Γ is measured from behaviour, not from the beliefs. For *policy* it matters: if the optimism is functional,

information interventions can destroy an asset while fixing a forecast, one more reason the chapter’s welfare claims never rest on belief correction alone. *The social arbitrage the triangle implies.* If people read everyone’s bias but their own, there is a comparative advantage waiting to be traded: I can police your temptations at low cost (yours do not tempt me) and you can police mine — which is arguably what study partners, held phones, and handed-over cigarette packs implement informally, and what peer-group commitment devices (Battaglini–Bénabou–Tirole’s peer self-control; savings monitors in the microfinance literature) implement formally. The triangle makes a sharp prediction for such designs, and the same data let us size it (Figure 2.13). Reading present-bias factors off Fedyk’s panel, actual behaviour sits at $\beta = 0.74$, belief about *others* at 0.91 (an error of .17 against the truth), belief about *self* at 0.97 (an error of .22): the *monitoring advantage* — how much more accurately people model others than themselves — is +0.05 in β units. That gap has teeth: a naïf setting her *own* commitment target aims at $\beta = 0.97$ (too lax to bind), whereas a partner using his accurate read of her aims at $\beta = 0.91$; delegating the choice to the partner closes 23% [2, 49] of the naïveté gap between self-belief and truth. This is a belief-based bound, so it isolates the *sophistication* channel (the partner knows your β) from the distinct *no-shared-temptation* channel (the partner simply isn’t tempted by your cigarettes) that also makes held phones and study partners work. *Mutual* commitment should therefore beat *self*-commitment wherever the first channel operates — a sharp, testable prediction for a field design (randomly paired vs. solo commitment), flagged for the sequel and now quantified rather than merely conjectured.

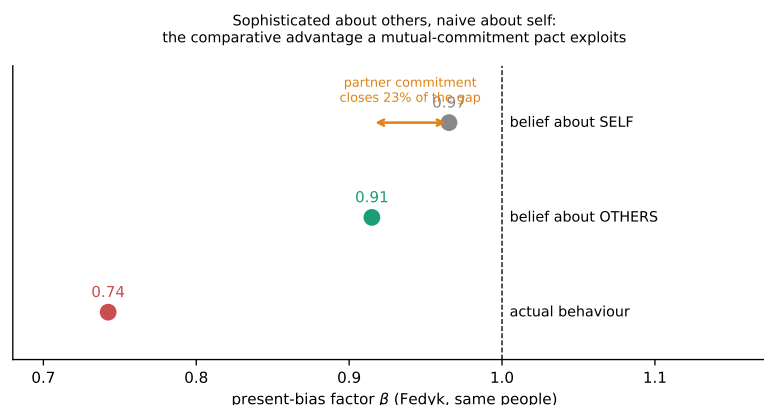


Figure 2.13: Mutual commitment, sized from the sophistication triangle (Fedyk). People read others’ present bias ($\beta = 0.91$, near the truth 0.74) far better than their own (0.97, essentially time-consistent). Delegating each person’s commitment to a partner who perceives them accurately closes 23% of the naïveté gap — the comparative advantage a mutual-commitment pact exploits, net of the separate no-shared-temptation channel.

The extension arms turn that lab triangle into a population fact (Figure 2.15, left panel). Wherever a dataset states both a plan (or a probability) and the outcome it refers to, the plan wins: New-Year resolvers state beliefs 11 points above their eventual self-rated success (Oscarsson, $n = 980$; overconfidence rejected jointly across belief quartiles, Cox–Shi $p < 10^{-4}$); gym members set four-week goals 4.6 visits above what the gym’s own turnstiles record — and 7.8 visits above their *own past month* — while stating a 63% chance of hitting the goal against a realized 25.5% and an 80.5% chance of meeting a 12-visit contract against a realized 42.2% (Carrera, both gaps +.37/+.38, Cox–Shi $p < 10^{-4}$); 12,452 HRS respondents in their fifties state a 52% probability of working full-time past 62 against a realized 42% (Section 2.14.7); and on Allcott–Gentzkow–Song’s phone-usage experiment — incentivized predictions against *app-measured* behaviour — untreated users underpredict their own screen time by 7.4 minutes/day three weeks out, 10.1 at six weeks, and 10.3 at nine (CS and ARP $p < 10^{-4}$, $n = 589$). The same

gym data adds the *memory* channel measured directly: self-reported past-100-day attendance exceeds the recorded count by 8.4 visits — plans are anchored on an inflated recollection of the past self, not only on an optimistic forecast of the future one.

The AGS predictions also settle two questions about the wedge’s *geometry* that no other dataset can (Figure 2.14). First, misprediction *grows with distance* on the digital margin: the nine-week wedge exceeds the three-week wedge by 2.8 min/day (se .9) within person — the A&R misprediction gradient, at field scale, against an objective target. Second, the fixed-target design (the same person predicts the same period at 6, 3, and 0 weeks out) shows the error barely shrinking as the target approaches (+9.5 $\rightarrow$ +10.4 $\rightarrow$ +8.3): people do not converge on themselves even at point-blank range. Life-cycle plans show the complementary geometry: the HRS wedge is *flat* in horizon (slope +.0005/year, +12pp even one year out) — persistent optimism, not distance-driven forecasting error. And beliefs repair slowly everywhere, at a rate that scales with feedback quality: resolvers absorb only .25 of each month’s surprise into the next belief (person-clustered [.20, .31]), phone users — who see their own dashboard — absorb .63 (se .03), and the residual stickiness survives even a randomized reward for accurate prediction (the |wedge| moves by +0.2 min/day [−3.1, +3.6] under a $5\times$ accuracy stake): naïveté is not insincerity and not ignorance, but slow learning about oneself.

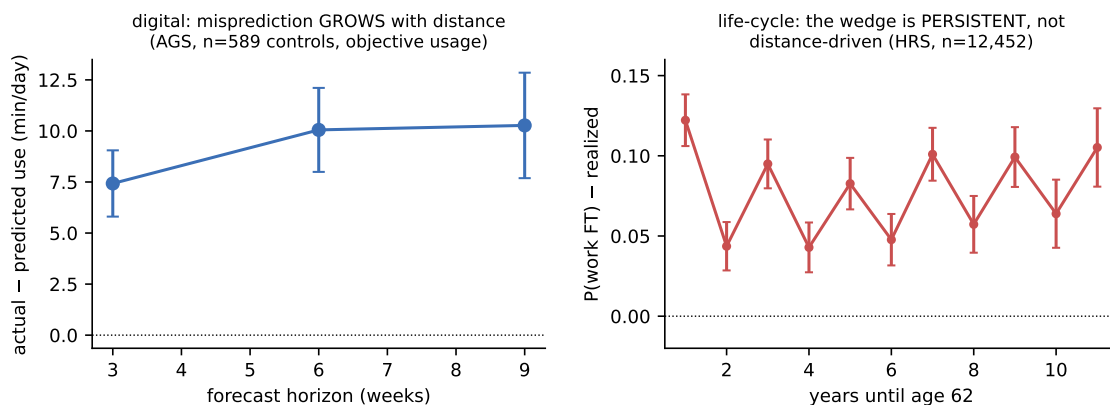


Figure 2.14: Two geometries of misprediction. Left: on the digital margin the wedge between app-measured and predicted own usage grows with the forecast horizon (AGS controls; incentivized predictions). Right: on the life-cycle margin the retirement-work wedge is persistent and flat in horizon (HRS) — optimism maintained by slow updating, not misforecasting at a distance.

Nothing available certifies the wedge (Figure 2.15, right panel). Commitment demand — the literature’s favourite revealed-preference witness for sophistication — correlates with the taker’s own wedge at +.02 (ANS), −.02 (Carrera, against overconfidence about the very contract chosen), +.11 (Carrera, against the goal wedge), −.18 (Sadoff, where takers have *smaller* wedges), +.12 with a $\pm .29$ interval (KKM, dominated-contract demand against the worker’s own payday gradient), and +.04 (AGS, willingness to pay for the app-limit tool against the user’s own measured prediction wedge — and +.03 against her own *stated* ideal reduction, +.00 against the addiction index: the valuation tracks neither the bias nor even the self-diagnosis); SEED’s hyperbolic flag raises commitment-savings take-up by a marginal +.17 among women and +.07 overall. Six datasets, both signs, nothing above .2: commitment take-up measures optimism about commitment, not knowledge of the bias. Choice-architecture experts do no better: on the StepUp megastudy’s 53 forecastable arms, behavioural practitioners’ predicted effects correlate −.24 with realized effects and overstate them eightfold on average (laypeople: +.10 and seventeenfold) — the beliefs of the people who *design* commitment devices are themselves miscalibrated about what the devices do. Two design consequences follow, now

resting on far more than one lab panel. *Never rest a welfare claim on commitment take-up or on anyone's stated beliefs. The welfare wedge Γ needs no sophistication assumption*: it is identified from plan-vs-realized comparisons at the advance vantage (where β cancels), for naïfs and sophisticates alike — and the extension arms show it is measurable at the scale of thousands.

Chosen deadlines close the loop between demand and efficacy: in the preregistered Ariely–Wertenbroch replication, 76% of subjects pay for commitment by setting interior deadlines, yet completion under self-set deadlines is *no better* than under no deadlines at all ($-.115$, $t = -1.0$), imposed even spacing does not beat freedom either ($-.017$, $t = -.2$), and within the self-set arm the earlier-deadline choosers do marginally *worse*. Commitment is demanded, paid for, and — on this margin — inert: exactly the pattern a naïf with sticky beliefs generates, and one more reason the welfare object is Γ , not take-up.

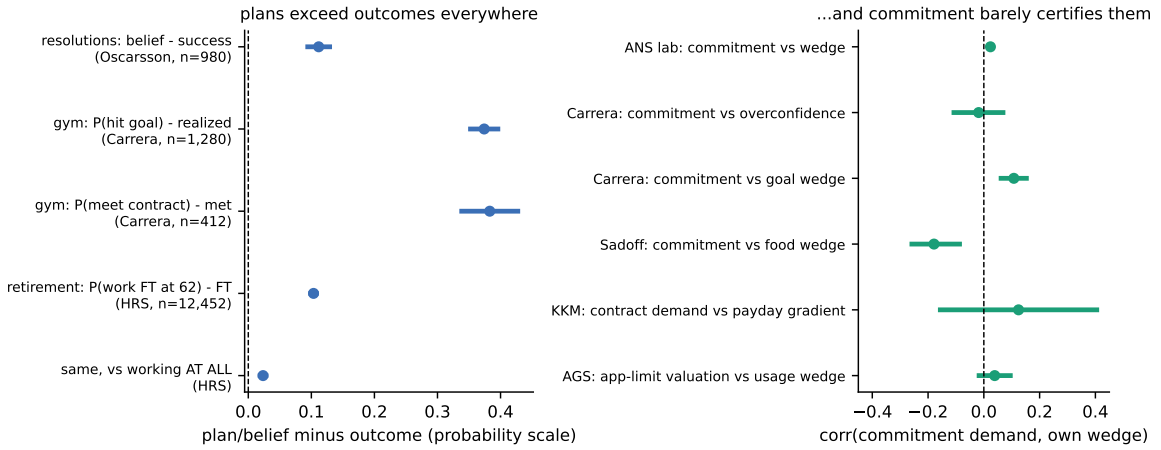


Figure 2.15: Left: plan/belief-minus-outcome wedges on the probability scale, with 95% CIs — resolutions (Oscarsson), gym goals and commitment beliefs (Carrera), and retirement work plans (HRS; the second row re-scores the plan against working at all, isolating the full-time margin). Right: the certifier panel — correlations between commitment demand and the taker's own wedge across four datasets plus the ANS lab benchmark. Plans exceed outcomes everywhere; commitment demand tracks none of it.

2.14.5 The symmetry audit, executed: one β cannot serve every choice

The trait-vs-state margins above test one family of (β, δ) restrictions (constancy across *states*); the taxonomy of Section 2.4 points at a second family (symmetry across the *sign, size, and domain* of the outcome), and the same effort data settle the sharpest of them. Augenblick–Niederle–Sprenger (2015) elicited monetary *and* real-effort discounting from the same 101 people; on their exact data, with the scale-free present-bias factor put on one scale (a two-limit Tobit handles the censored effort corners), effort is present-biased while money is essentially not: $\beta_{\text{effort}} = 0.88$ vs. $\beta_{\text{money}} = 0.99$ (Figure 2.16). The gap is $+0.10$ $[0.04, 0.19]$ within person, and a subject bootstrap rejects a *single* present-bias factor at $p < .001$. The asymmetry is not an artefact of how the corners are treated: re-estimating the money factor with the *same* two-limit Tobit used for the censored effort corners leaves it at 0.985 (OLS 0.985), so a common censoring-aware estimator preserves the gap. Known: ANS report the pooled money–effort difference. New here: the within-person present-bias-factor distribution, and — this is the point — reading it as one row of a systematic audit rather than an isolated curiosity. Put beside the stakes trend (Section 2.14.3), the proximity gradient, and the anticipation reversal of Section 2.9, it completes the verdict of Table 2.1: four of the five (β, δ) symmetries are rejected, each by the contrast the taxonomy assigns it. *This is the chapter's cleanest demonstration that β – δ is not merely imprecise but structurally too symmetric*: it forces one number onto choices that

the person herself treats differently, and MSC — distortion on the hot immediate component, which is an approach-system reward for money-like temptations and an avoidance-system cost for effort — predicts exactly the pattern of breaks, and only that pattern. The one symmetry we cannot yet isolate is sign *within* a domain (gains vs. losses of the same good), because the money–effort contrast moves domain and sign together; a same-good gain/loss design is the natural sequel.

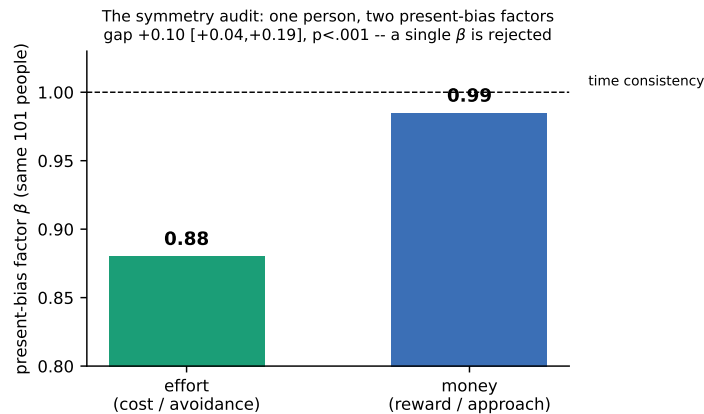


Figure 2.16: The symmetry audit’s sharpest row: the same 101 people carry two different present-bias factors — effort (a cost, the avoidance system) is present-biased at $\beta = 0.88$, money (a reward, the approach system) is essentially time-consistent at $\beta = 0.99$. A single (β, δ) number is rejected within person ($p < .001$); the wedge lives where there is a hot immediate component, as MSC predicts.

2.14.6 The population answer: type shares

Classifying each Cerrone participant by three margins (effort wedge, money wedge, sign-sophistication) — with thresholds and a design-simulated mixing matrix calibrated to the *within-subject sampling error* of each margin, so genuine heterogeneity stays attributable to types — the LP delivers set-identified shares, feasible at the nominal level: **time-consistent** [0.67, 0.79]; **genuinely present-biased naïfs** [0.09, 0.32] — bounded away from zero — **sophisticated present-biased** [0.00, 0.11]; **future-biased** [0.00, 0.02]; **money-driven** [0.00, 0.12]. The aggregate wedge is real and is carried by a present-biased minority of roughly one person in ten to one in three, mostly naïve about it.

The extension data replicate the minority structure on other margins and add a warning about mislabeling. In the grocery panel, 35.1% of people switch toward more calories when the choice turns immediate and 17.1% switch the other way — a present-biased plurality over a future-biased minority, with half the panel time-consistent, on real food. In Halevy’s two-session design the three margins the literature blurs come apart person by person: 61% are stationary, 50% time-consistent, 42% time-invariant — and crucially 29% are *stationary but inconsistent* while 16% are *consistent but non-stationary*, so a stationarity test used as a consistency test (which is what a one-shot MEL experiment is) misclassifies almost half the sample. And at MOOC scale the declared-intent gradient is informative but level-inflated: realized engagement and completion rise monotonically in stated expected hours (completion 6.5% → 24.6% across intent bins, 36,803 stated intents), yet a quarter of the highest-intent students are active less than one day a week — stated intentions rank people correctly and still overstate everyone. This is the “how much is genuine” question answered the only way it can be answered honestly: as population shares with set identification, not as one β for everyone.

2.14.7 The field arm: consumption, work, exercise, and life-cycle plans

On Aguiar–Kashaev’s exact ECPF input (185 singles, 4 quarters, 17 goods, quarterly interest rates), the deterministic Browning/life-cycle test rejects exponential discounting for **81.6% of singles** and 88.5% of couples — reproducing the deterministic benchmark. Read against the Monte-Carlo’s certificate (83% *spurious* rejection under 10% measurement error), those numbers are upper bounds on irrationality, not verdicts. The noise-robust verdict is the (β, δ) confidence set (Figure 2.17): exponential discounting fits *perfectly* ($TS = 0.0$, reproducing Aguiar–Kashaev’s conclusion with our independent implementation); $\beta = .90\text{--}.95$ are accepted at every δ ; $\beta = .80\text{--}.85$ survive only by compensating with $\delta = 1$; and $\beta \leq .75$ is **rejected outright** ($TS = 49$ at $.70$). A 400-household couples subsample gives the *same* set — even though deterministic tests reject 89% of couples — so the aggregation excess visible in feasibility shares does not survive noise-robust inference, exactly as the mixture-of-deltas arithmetic (T7c) predicts.

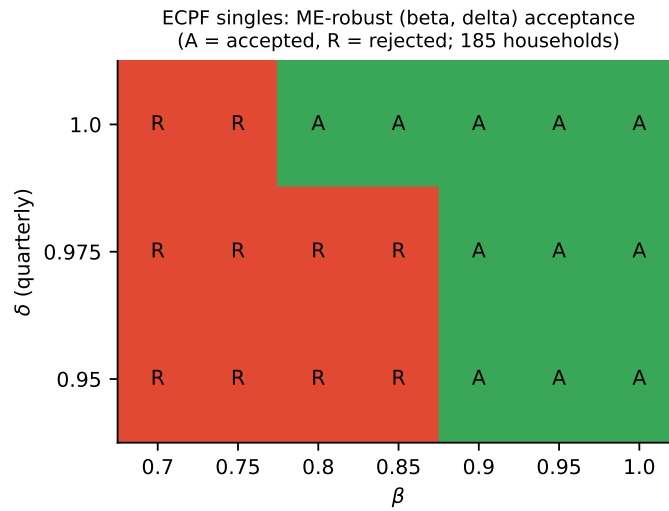


Figure 2.17: The ECPF singles (β, δ) acceptance grid (A = accepted, R = rejected; ELVIS-RP with centered measurement error, conservative χ^2_8). Deep present bias is excluded; time consistency fits exactly; mildly present-biased points survive only with $\delta = 1$. The couples subsample gives the same set.

Consumption is one field margin; the extension arm adds five more, each with its own instrument. **Exercise, with 52 randomized instruments at once** (Figure 2.18). On the StepUp megastudy’s 61,293 gym members the base planning-plus-micro-incentives program lifts the weekly visit probability by 1.6pp over placebo during the four incentivized weeks, the best arms by 3.4pp — and doubling the micro-incentive adds only +0.6pp $[-1.0, +2.1]$ on top of the base effect — a reward-salience response, small and concave (the points are near-immediate, so this complements rather than tests the $\beta(\text{stakes})$ question). The new object is *persistence*: once incentives stop, equal decay across mechanisms is rejected (Wald $p = .016$). Temptation bundling — the free audiobook playable only at the gym — retains 89% of its during-program effect ten weeks out; higher incentives retain 70%; planning add-ons 27%; pledges 4%; framing goes negative. “Habit formation” is not a mechanical stock of repetitions: *what* built the behaviour determines what survives its removal, exactly as a temptation-cost account (make the gym itself less aversive) and against a pure repetition account. **Contract choice as a revealed-preference bound.** DellaVigna–Malmendier’s monthly members pay \$15.8 per visit in aggregate against a \$10 pay-per-visit alternative; rationalizing the monthly contract requires expected attendance of at least fee/10, so fee/10 – attendance is an assumption-light

lower bound on the plan-vs-outcome wedge: +2.6 visits/month on average, positive for 78% of members, with 9.7 months on average between the last visit and the effective end of the contract — overprediction or switching costs, and the paper claims only the disjunction. **Life-cycle plans.** In the HRS, the stated probability of working full-time past 62 exceeds the realized rate by 10.4pp [9.5, 11.2] ($n = 12,452$; overstatement rejected jointly across belief bins, Cox–Shi $p < 10^{-4}$; the top bin states 99.7 and realizes 64.3). The decomposition is the point: re-scoring the plan against working *at all* shrinks the wedge to +2.3pp — most of the miss is the full-time margin, a downshift rather than an exit — while intervening health shocks, though real ($t \approx 6\text{--}13$), explain only 0.6% of the individual variance. And the wedge does *not* grow with horizon (slope +.0005/year; it is +12pp even one year out), while within-person revisions drift *upward* (+1.6 points per wave against a martingale’s zero): at life scale the plan-outcome gap is not distance-driven misprediction but a persistent, slowly-corrected self-image — the same sticky updating the resolution panel shows month by month. **Learning.** The OULAD deadline hazards above complete the set: 29.5% of 167,044 submissions are late, a third arrive within 24 hours of the deadline, and the most-deprived decile is 4.1pp later than the least-deprived — the population margin, descriptive and stated as such. extbfDigital addiction — the cue-rich margin, with two randomized instruments. On Allcott–Gentzkow–Song’s experiment (1,933 Android users, app-measured usage, our analysis of their public microdata) the randomized \$2.50/hour bonus cuts FITSBY screen time by 63 min/day while active and by 17 and 11 min/day in the two post-bonus periods — 27% and 17% persistence, the habit stock their structural model needs, reproduced model-free — and the randomized limit tool delivers a stable -23 to -19 min/day for nine weeks. The chapter’s addition is the *belief geometry around those instruments*: treated users’ *predicted* treatment effects match their realized ones almost exactly ($z = .15$ and $.46$ in the post periods) even while the same users mispredict their usage *levels* by 7–10 min/day — people are sophisticated about how incentives will move them and naïve about where they stand, which is precisely the pattern that makes level-based self-regulation fail while leaving demand for incentives intact. The addiction-scale gradient of the wedge is +4.9 min/day $[-0.6, +10.4]$ — suggestive, not conclusive — and this experiment’s certifier and horizon results appear in Section efsec:beliefresults.

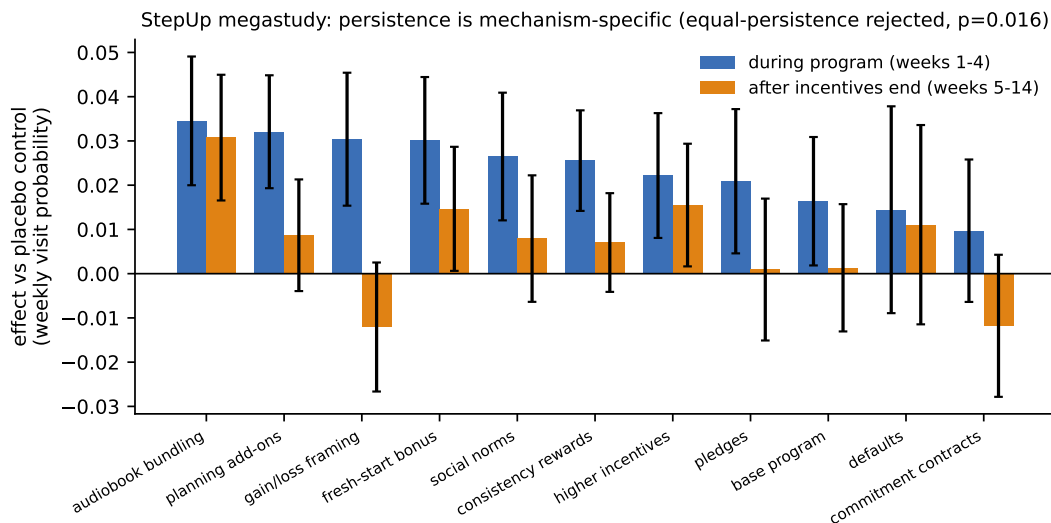


Figure 2.18: The StepUp megastudy through the decomposition lens: per-family effects on the weekly visit probability versus placebo control, during the four incentivized weeks (blue) and in the ten weeks after incentives end (orange); 95% CIs, person-level randomization. Equal persistence across mechanisms is rejected ($p = .016$): temptation bundling persists, pledges and framing do not — habit is mechanism-specific, not a stock of repetitions.

Read together, the arms are the paper’s sharpest sentence: *over effort, the set of accepted models excludes time consistency* ($[0.75, 0.95]$ and $[0.80, 0.95]$ in two independent panels); *over household consumption, it excludes deep present bias* ($[0.80, 1.00]$) — domain-specificity established noise-robustly on both sides, with no functional forms and nobody dropped, and with the overlap $[0.80, 0.95]$ identifying the only β values a *unified* account could still defend. Around that core, the same machinery now delivers, in the field: plan-vs-outcome wedges of +10 to +38 percentage points wherever plans are stated; effort spikes that track deadlines and paydays but not stakes; commitment that is demanded and paid for yet certifies nothing and (on the deadline margin) improves nothing; and persistence that depends on the mechanism that built the behaviour.

What the code taught the design (lessons that now bind)

(i) Projection bias reproduces the full state gradient; only *matched-state advance decisions* separate it from a true $\beta_{\text{eff}}(X)$ — a hard requirement for the cue/load micro-survey and a stated qualifier on every unmatched state gradient. (ii) In compact multi-day designs, money-immediacy is a function of the decision day: β_m is not separately identified from day effects (Cerrone’s money parameter is absorbed by day fixed effects; their effort parameter survives at +.41). Day-fixed-effect robustness is a standard row. (iii) Commitment demand is uncorrelated with the wedge it certifies; welfare rests on Γ , which the sophistication triangle shows needs no sophistication assumption. (iv) Temptation and fixed action costs are observationally identical on the stakes margin; the paper claims only the family split (multiplicative vs. additive), which the data settle for multiplicative. (v) Determinism is not rigor: without price variation the deterministic test cannot see β ; with noise it manufactures irrationality — every feasibility number is paired with its noise-adjusted reading. (vi) Quadratic-form inequality tests need non-redundant moments: ordering nulls are encoded by adjacent increments. (vii) Censored choices mechanically violate centered-ME moments: corner-heavy panels are analysed on interior units, with the all-unit rejection reported as the certified artefact it is. (viii) The welfare wedge Γ must be *measured*, never proxied: plans exceed outcomes in every dataset that states both (from +10pp on retirement plans to +38pp on gym contracts), beliefs absorb only a quarter of each period’s surprise, recollection of the past self is inflated, and commitment demand correlates with the taker’s own wedge at $|r| \leq .18$ with unstable sign across six datasets (including the digital-addiction margin, where it tracks neither the measured wedge nor the stated ideal) — neither stated beliefs nor take-up can stand in for Γ . (ix) Post-intervention behaviour is mechanism-specific (equal persistence across the megastudy’s arm families is rejected, $p = .016$; temptation bundling retains 89%, pledges 4%): no welfare extrapolation from “the habit formed” without naming what formed it.

Essential readings — The essential readings for this chapter, in priority order

Read these six first, in this order — they carry the argument end to end: (1) Ericson–Laibson (2019, Handbook of BE) — the identification map (commitment $\times$ consistency) the chapter is organised around; (2) Augenblick–Niederle–Sprenger (2015, QJE) — the effort-discounting design and the money-vs-effort contrast that becomes the symmetry audit; (3) Augenblick–Rabin (2019, REStud) — multi-vantage wedges, sophistication vs projection; (4) Blow–Browning–Crawford (2021, REStud) — the nonparametric time-RP method (the nearest competitor); (5) Fudenberg–Levine (2006, AER) — the dual-self model MSC extends and is tested against; (6) Bernheim–Rangel (2009, QJE) — welfare without rationality, the Γ backbone.

Then, by role:

- **Backbone (methods):** Halevy (2015, Ecma); Aguiar–Kashaev (2021, REStud); Andrews–Roth–Pakes (2023) and Cox–Shi (2023, REStud); Beatty–Crawford (2011, AER); Schennach (2014, Ecma).
- **Confounds:** Blow–Browning–Crawford (2021, REStud) and Crawford (2010); Andreoni–Sprenger (2012b) with Epper et al. (2011) and Halevy (2008); Dean–Sautmann (2021), Cassidy (2019), Epper (2017); Balakrishnan et al. (2020); Adams et al. (2014) and Jackson–Yariv (2014/2015) with Millner–Heal (2018); Benhabib–Bisin–Schotter (2010).
- **Inference:** Schennach (2014, Ecma); Andrews–Roth–Pakes (2023, REStud) and Cox–Shi (2023, REStud); Beatty–Crawford (2011, AER); Varian (1985); Browning (1989).
- **Model ladder:** Laibson (1997); O’Donoghue–Rabin (1999, 2001); Fudenberg–Levine (2006); Gul–Pesendorfer (2001); Andersen et al. (2008); Cohen et al. (2020) and Cheung–Tymula–Wang (2021).
- **MSC scaffold, and its evolutionary microfoundation:** the dual-self and control layer — Fudenberg–Levine (2006); Loewenstein–O’Donoghue (2005); Brocas–Carrillo (2005/2021); Shenhav–Botvinick–Cohen (2013); Daw–Niv–Dayan (2005); Kurtz–David et al. (2019); Ainslie (1992), Schelling (1984), Frankfurt (1971) — and the mismatch microfoundation new to this draft: Hayden (2018) on why evolution left self-control imperfect; Burnham–Phelan (2020) on mismatch and the superstimulus; Wertenbroch et al. (2008) and Tao–Chen (2023) on the older (Adam-and-Eve) framing of temptation; with Bernheim–Rangel (2009) for welfare.
- **Anticipation (Test B):** Loewenstein (1987); Caplin–Leahy (2001); Ganguly–Tasoff (2016); Falk–Zimmermann (2024).

Chapter 3

Rationality under Risk: Expected Utility vs. Probability Weighting

3.1 The theory

3.1.1 Expected utility and subjective expected utility

Under **expected utility** (EU; von Neumann–Morgenstern), an agent ranks risky prospects by $\sum_s \pi_s u(x_s)$, where π_s is the (objective) probability of state s and u is a Bernoulli utility over outcomes; risk aversion is *equivalent to concavity of u* . Under **subjective expected utility** (SEU; Savage), probabilities are not given but are a *subjective belief* μ_s inferred from choices, and the agent maximizes $\sum_s \mu_s u(x_s)$. These are the workhorses of finance, insurance, and macro.

In a market/portfolio setting — the one that matters for revealed-preference testing — the agent allocates a budget across *state-contingent claims* (an amount x_s delivered if state s occurs) at state prices p_s . The first-order condition is $\mu_s u'(x_s)/[\mu_{s'} u'(x_{s'})] = p_s/p_{s'}$: a risk-averse SEU agent buys *more* consumption in states that are cheaper or more likely. This is the “inverse price–quantity” signature of Chapter 1, specialized to risk.

Intuition — Why “concave utility” means “risk averse,” in one example

Utility u over money is **concave** when each extra dollar adds less happiness than the one before (diminishing marginal utility). Consider a 50–50 bet to win or lose \$100 around wealth \$1,000. The \$100 you might *lose* costs you more utility than the \$100 you might *win* adds — the dollars you would give up are “worth more” than the dollars you would gain. So the fair bet has negative expected utility and you decline it: that *is* risk aversion. Expected utility ties your whole attitude to risk to the curvature of u — which (next) is also its fatal weakness.

3.1.2 The sharpest vulnerability: Rabin’s theorem

Theorem 3.1 (Rabin 2000, calibration). *If an expected-utility maximizer with concave u rejects a small, actuarially favourable gamble (e.g. a 50–50 chance to lose \$100 or gain \$110) at every wealth level in a range, then she must reject enormously favourable large gambles (e.g. 50–50 to lose \$1,000 or gain any sum). The result is fully nonparametric (only concavity is used).*

Because real people *do* refuse modest favourable bets yet would obviously accept huge favourable ones, *any* non-trivial small-stakes risk aversion is a *logical refutation* of EU. Rabin himself concludes that experimental methods relying on EU’s small-stakes risk-neutrality “should be abandoned.” This hands you a deductive lever, not just an anomaly to document.

Intuition — Why Rabin’s theorem bites

If you reject a 50–50 lose-\$100 / gain-\$110 bet at *every* wealth level, expected utility can only explain it by marginal utility falling *fast*: the dollars just above your wealth must be worth noticeably less than the dollars just below. But if that holds at *every* wealth level, marginal utility keeps shrinking at that rate as you get richer — compounding until, a few thousand dollars up, extra money is worth almost nothing. You would then refuse even a 50–50 bet to lose \$1,000 or gain *millions*, which nobody would. So a modest, sensible refusal, taken literally under EU, *forces* an absurd one — meaning EU is the wrong account of small-stakes risk aversion, and something like probability weighting is needed.

3.1.3 The behavioral rival: prospect theory and probability weighting

Kahneman–Tversky (1979) replace EU with two precise departures: (i) a **value function** over *gains and losses relative to a reference point*, concave for gains and convex for losses (diminishing sensitivity) and *steeper for losses* (**loss aversion**, roughly twofold); and (ii) **decision weights** $\pi(p)$ that replace probabilities, *overweighting small probabilities* and underweighting moderate ones, with $\pi(p) + \pi(1 - p) < 1$. The rank-dependent formulation (Quiggin 1982) makes this coherent over many outcomes; disappointment aversion (Gul 1991) and maxmin EU (Gilboa–Schmeidler 1989, for ambiguity) are further alternatives.

Friedman–Savage, resolved by weighting — the field test you will run

The same household *insures* (risk-averse) *and gambles* (risk-loving). No single concave (or convex) u explains both — the Friedman–Savage (1948) puzzle. But probability weighting does: Kahneman and Tversky themselves note that overweighting small probabilities “may contribute to the attractiveness of *both* insurance and gambling.” Hence the within-household *co-existence of insuring and gambling* is a direct, field-observable discriminator between EU and probability weighting.

3.2 Testable predictions, as inequalities

The RP characterizations specialize the Afriat machinery to risk. Echenique–Saito (2015) give necessary-and-sufficient conditions for (risk-averse) SEU from portfolio data (using “doubly balanced” price-quantity sequences and the 45° line). Polisson–Quah–Renou (2020) give a general nonparametric method — “GRID” — that tests EU, rank-dependent utility, disappointment aversion, and maxmin by restricting the infinite consumption space to a finite grid generated by the data, turning each theory into finitely many linear inequalities, and *crucially does not require concavity*, so it admits *risk-loving* (gambling) behaviour. GRID also yields an Afriat-style efficiency index for the magnitude of violations.

3.3 Existing tests, and what they assume

1. **Lab revealed preference.** Choi, Fisman, Gale & Kariv (2007) and Choi, Kariv, Müller & Silverman (2014) have thousands of subjects choose portfolios of state-contingent claims from budgets; they score GARP/CCEI and show consistency *predicts real-world wealth* (a one-SD increase in consistency $\approx$ 15–19% more wealth). Echenique–Imai–Saito (2019) find most subjects *are* utility maximizers but the *majority violate SEU*, with maxmin adding almost nothing. *Lab, artificial stakes; the authors note field evidence is missing.*
2. **Field, structural.** Barseghyan, Molinari, O’Donoghue & Teitelbaum (2013) estimate a model from *insurance deductible* choices and find **probability distortions** (overweighting

small claim probabilities) dominate — not standard risk aversion, and not loss aversion or disappointment aversion alone. *Parametric/semi-nonparametric, insurance-only, observational (no natural experiment); inconsistent choices absorbed as logit noise.*

3. **Field, betting.** Andrikogiannopoulou (2011) fits parametric prospect theory to *online-sportsbook* data and recovers probability weighting — so “betting + risk preferences” is not virgin territory. *Parametric, betting-only, no natural experiment, not nonparametric RP.*
4. **The method, in the lab only.** Polisson–Quah–Renou (2020) apply GRID to lab portfolio experiments. They *explicitly note* (fn. 13) the method applies to “observational budgetary choice data of the type found in *insurance or financial decision problems*,” but do not run it there.
5. **The natural experiment, reduced form.** Baker et al. (2024) and Hollenbeck et al. (2024) use the staggered U.S. legalization of online sports betting to show it crowds out saving and worsens credit. *They estimate financial harm, not a test of EU.*

3.4 The gap

Field risk preferences are studied *structurally* and *insurance-only* (Barseghyan et al.) or *betting-only* and *parametric* (Andrikogiannopoulou); the nonparametric method (GRID) has been used only in the lab; the betting natural experiment has been used only for reduced-form harm. *No one runs a nonparametric RP test of EU vs. probability weighting in the field, identified by a natural experiment, on the insure-and-gamble co-existence.* That intersection is open — and the authors of the method point straight at it.

3.5 An executed instance: EU rejected by a certainty premium (lab, run now)

The field design below is the frontier; but the same machine can be run *today* on real, randomized data, and doing so both proves the method and feeds back into Chapter 2. Andreoni–Sprenger’s (2012b) “Risk Preferences Are Not Time Preferences” is a convex time budget in which each of two dated accounts pays with a *randomized probability*. Six probability conditions are crossed so that the payment-probability *ratio* π_1/π_2 is matched across a certain-containing condition and an all-risky one: (100, 80) vs. (50, 40) (ratio 1.25, certainty on the sooner account), (80, 100) vs. (40, 50) (ratio 0.8, certainty on the later account), and (100, 100) vs. (50, 50) (ratio 1, symmetric certainty). This randomized probability *is* the instrument Z the framework asks for.

The EU restriction, and the certainty premium that violates it

With expected utility $\pi_1 u(c_1) + \pi_2 \delta u(c_2)$ and a CRRA u , the first-order condition gives $\log(c_1/c_2) = -\frac{1}{\rho} [\log p - \log(\pi_1/\pi_2)]$: the probabilities enter *only* through the ratio π_1/π_2 and with the *same* coefficient as the time-price p . So within a matched probability ratio the token allocation must be *identical* whether or not one account is certain. A systematic shift toward the certain account — a *certainty premium* — is a common-ratio (Allais) violation, the revealed-preference signature of probability weighting. It is also exactly the object Chapter 2’s “implicit-risk” masquerade (channel 2) predicts: it is what inflates apparent present bias when the immediate account is certain and the future one is not.

What A&S do, and what we add. A&S fit a parametric CRRA-with-curvature model by nonlinear least squares, drop non-monotone subjects, and read the premium off a structural parameter. We instead (i) keep all 80 subjects and all corner allocations (33.8% at $c_1=0$, 14.3%

at $c_2=0$) with a two-limit Tobit; (ii) test the EU restriction *nonparametrically* as a moment inequality on within-subject matched wedges (Cox–Shi and ARP); (iii) *condition on stakes* (the gross return) and horizon in quantile bins, asking whether the premium holds in *every* bin, not on average; and (iv) run a placebo — the symmetric-certainty ratio-1 pair — that must show no asymmetric premium if the effect is genuine certainty overweighting rather than a risk-level artifact.

The verdict: EU is rejected, cleanly. Making the sooner account certain pulls +4.4 tokens toward it; making the later account certain pulls 5.0 tokens away from the sooner ($z = 6.7$ and 6.9 ; EU forbids both). Probabilities enter the allocation with a *smaller* coefficient than the matched price (-10.3 vs. -27.1 ; equality $z = -6.8$) — the flattening a weighting function produces. The within-subject matched certainty premium is +2.67 tokens $[2.06, 3.29]$, with Cox–Shi and ARP $p < 10^{-15}$; it is positive in *every* stake quartile (2.05, 2.98, 3.30, 2.35) and at both horizons (+2.74 at 28 days, +2.60 at 56), while the symmetric-certainty placebo carries no positive premium (Figure 3.1). This is the nonparametric RP test of EU vs. probability weighting, censoring-aware and dropping nobody, executed on real randomized data — and it lands the same verdict A&S reach parametrically, without their functional form or their sample cuts.

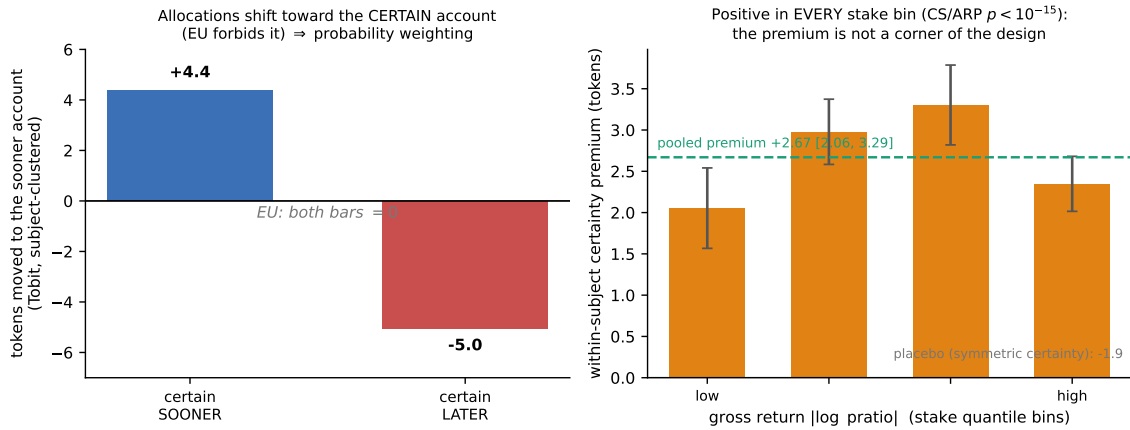


Figure 3.1: The executed risk arm (Andreoni–Sprenger 2012b). *Left*: allocations shift toward whichever account is *certain* — +4.4 tokens to a certain sooner account, -5.0 to the sooner when the later is certain — which expected utility forbids (both should be zero), so probabilities are weighted, not linear. *Right*: the within-subject certainty premium is positive in every stake bin (Cox–Shi/ARP $p < 10^{-15}$) and the symmetric-certainty placebo is null, isolating it as the asymmetric certainty effect. The same premium is what masquerades as monetary present bias in Chapter 2 — one reason $\beta_{\text{money}} \approx 1$ there once the certainty of the immediate payment is stripped out.

The point is not that this lab result settles field behaviour — it does not, and the gambling/insurance design below is where the stakes and the external validity are. The point is that the *method* is not hypothetical: given randomized probabilities, the EU restriction becomes a handful of moment inequalities that real data can and do reject, keeping every subject and every corner. The field test scales exactly this logic to a natural experiment.

3.6 Your new test

How to go beyond — Nonparametric EU testing in the field, via the betting boom

(1) Theory as inequalities. Use Polisson–Quah–Renou’s GRID (it admits risk-loving, so it can handle gambling) to test EU against rank-dependent / probability-weighting alternatives with no functional form; use Echenique–Saito for the SEU layer.

(2) The instrument Z . Use *staggered state legalization* of online sports betting (post-*Murphy v. NCAA*, 2018) as exogenous variation in the price/availability of risk-taking. Legalization timing is driven by legislative idiosyncrasies, plausibly orthogonal to individual risk preferences — the GARP-conditional-on- Z design, here in the field.

(3) The Friedman–Savage co-existence. Link betting to insurance/financial data and test EU on the *same households*’ insuring *and* gambling — the sharpest discriminator between EU and probability weighting — with Rabin’s theorem as a deductive bound (any non-trivial small-stakes aversion among the same agents already strains EU).

(4) Bin over Z . Test the GRID inequalities within bins defined by legalization status/intensity and time-since-legalization, and test whether the recovered preference class (EU vs. rank-dependent) is stable across bins; this turns the conditional restriction into ARP-testable unconditional ones.

(5) Inference and power. ARP moment-inequality inference; Bronars/efficiency-index power benchmarks; statistical RP for choice noise.

Data. Financial-aggregator transaction panels (the data Baker et al. use), linked to insurance choices; state legalization dates and rollout intensity.

Verdict (Type A/B). *Type B*: field EU fails in a characterizable direction (probability weighting / reference dependence), pinning the behavioral foundation for gambling regulation and for the distributive-justice debate. *Type A*: “gambling” is rationalizable (entertainment value; naivete that fades with experience) and field EU survives — overturning the lab pessimism.

What would falsify EU. A household that simultaneously reveals risk aversion (insurance, small-stakes rejection) and risk-loving (gambling) at prices that no single concave u can reconcile — detected nonparametrically by GRID, robust across Z -bins, and significant under ARP.

Intuition — Concretely: the inequalities, the moment, and what Z is here

The inequalities. GRID asks whether a household’s portfolio and insurance choices can be rationalized by *one* expected-utility function on a finite grid — again a system of *linear inequalities*, now with the EU probability-weighted-sum structure built in (with weaker variants for rank-dependent utility, which permit probability weighting). *What Z is.* Z is the *legalization of online betting*: the date and intensity with which a state switches betting on. It shifts the *availability and price of risk-taking* but is set by legislative timing, not by any individual’s risk tastes. *The moment / what is tested.* Within bins of Z (states/periods by legalization), test whether the *same* households’ insuring *and* gambling fit a single u . The EU inequalities fail exactly when risk-averse insuring and risk-loving gambling coexist at prices no concave u allows — a violation that, if real, should appear *systematically across* Z (tested with ARP). A robust failure points to probability weighting.

3.6.1 A worked example: how the test could be run (illustrative)

One concrete design, to fix ideas — not the only possible approach.

Data. A household-quarter panel linking (a) insurance choices (e.g. deductible menus — a choice among state-contingent payoffs) and (b) sports-betting activity (a bet is a state-contingent claim bought at odds that act as state prices). Together they form budgets over “money if the event happens / money if not” — the input GRID needs.

The instrument Z (here an ordered dose anchored on a clean switch). Begin from the exogenous binary switch $Z_{s,t}^0 = 1\{\text{online betting legal in state } s \text{ at } t\}$ — the cleanest part (legislative timing). Then enrich it to a dose: $Z_{s,t}$ = quarters-since-legalization, or licensed mobile sportsbooks per adult, to see whether *more* exposure pushes choices further from expected utility.

Step 1 — RP test per household. Run GRID: can a *single* expected-utility function rationalize this household’s insuring *and* betting on a finite grid? Record pass/fail and the Afriat efficiency (how badly EU fails).

Step 2 — bin on Z . Compare EU pass-rates and efficiency across Z -bins (not-yet-legal; 0–1 yr; 1–2 yr; 2+ yr legal, or low/medium/high intensity).

Step 3 — moment and inference. Test whether EU-violation severity *rises with Z* — does access to legal betting make the same households’ insure-and-gamble pattern progressively impossible to reconcile with one concave u ? — with Andrews–Roth–Pakes inference; Rabin’s calibration adds a deductive check on the insurance side.

Reading the result. A robust, Z -graded failure (risk-averse insuring *and* risk-loving betting at prices no concave u permits) is evidence for probability weighting over EU (Type B). If choices stay EU-rationalizable as betting becomes available, field EU survives (Type A).

Why it matters — Domain of Chapter 3

Gambling regulation; chronic under-saving for retirement and under-insurance (health, climate); social-insurance design. And a deep payoff: the foundations of distributive justice are partly empirical — Rawls’s difference principle *assumes* strong risk aversion, Harsanyi’s utilitarianism *assumes* EU, Dworkin’s “option luck” treats deliberate gambling as a fair source of inequality. How people actually treat risk speaks to which social-contract theory has realistic behavioral foundations.

Essential readings — Chapter 3 — rationality under risk

(1) Rabin (2000) — the calibration theorem (short, decisive). (2) Friedman & Savage (1948) — the puzzle. (3) Kahneman & Tversky (1979) — prospect theory (value function, weighting, the certainty/reflection/isolation effects). (4) Polisson, Quah & Renou (2020) — GRID (*read closely; this is your method*). (5) Echenique & Saito (2015) — the SEU revealed-preference characterization. (6) Barseghyan, Molinari, O’Donoghue & Teitelbaum (2013) — the field-risk foil; their (2018, *JEL*) survey for breadth. (7) Echenique, Imai & Saito (2019) — lab SEU violations. (8) Choi, Kariv, Müller & Silverman (2014) — large-scale RP and outcomes. (9) Barberis (2013); Köszegi & Rabin (2006) — the behavioral rival in depth. (10) Baker et al. (2024) — the betting natural experiment (data and design).

Chapter 4

Rationality of Machines: Can We Do Welfare Economics on What We Delegate to AI?

As economic decisions move to AI agents — pricing, lending, hiring, portfolio choice, procurement — two assumptions that all of welfare economics and market design rely on come into question. Both are testable with the *same* revealed-preference toolkit, and this chapter has two threads accordingly.

4.1 Thread 1 — Are AI agents rational, and is welfare analysis on their choices even defined?

4.1.1 The theory

The object is unchanged from Chapter 1: an AI agent’s choices are *rational* if they are rationalizable by a stable utility (Afriat/GARP), and we can ask the stronger questions too (SEU under risk, via Echenique–Saito). The reason this matters is sharper than “is the model smart?”:

The welfare stake (Bernheim–Rangel applied to delegation)

If an AI’s choices cannot be rationalized by stable preferences, then “consumer surplus” and willingness-to-pay are *undefined* for AI-mediated markets — we cannot evaluate them. Bernheim–Rangel (2009) make this precise: the AI’s *prompt/frame* is an **ancillary condition**, and their analysis of *delegation* (a choice-relevant condition changes meaning when the decision is delegated from a person to a planner — or to an AI) implies that *if the AI’s choices flip with payoff-irrelevant frames, the welfare-relevant domain is ambiguous*, and welfare analysis on delegated AI decisions is not well-defined. In Thaler’s language, the frame is a **supposedly irrelevant factor**; the question is whether the AI treats it as irrelevant, as rationality requires.

4.1.2 Existing work, and what it does *not* do

1. Chen, Liu, Shan & Zhong (2023, *PNAS*) had GPT make budget choices and found near-perfect GARP/CCEI (≈ 0.997 – 0.999 , “more rational than humans”). *But their own paper reports that rationality drops sharply under a different price framing and under discrete choice* — noted as a caveat.

2. **Raman et al. (2024, “STEER”)** build a large benchmark grading 14 models on 64 “elements of rationality” via multiple-choice accuracy — a “report card.” Related: GARP-EFM (2026) uses Afriat’s theorem to fine-tune models; “utility engineering” (Mazeika et al. 2025) studies emergent value systems; Horton (2023) proposes LLMs as simulated economic agents (“homo silicus”).

What none of these do: (i) revealed-preference axiom tests with *econometric inference* (they report scores, not hypothesis tests with correct size); (ii) *identification* — they document frame-sensitivity but do not treat it as a confounder to instrument out; (iii) the *welfare* question — whether evaluation of AI-mediated choices is defined; (iv) *exploitability* in a market.

4.1.3 Your new test (Thread 1)

How to go beyond — An econometric, welfare-relevant revealed-preference test of AI agents

(1) **Theory as inequalities.** Elicit AI choices from randomized budgets (as Chen et al., but as the input to a *test*, not a score); apply GARP/Afriat and the SEU layer.

(2) **The instrument Z = a randomized, payoff-irrelevant frame.** The AI’s choice depends on the prompt, which can correlate with the “prices” in the task — the same confounder as everywhere else. *Randomize a payoff-irrelevant frame* (wording, order, surface context) as Z and test whether the revealed preference is *invariant* to it. Invariance is what utility theory requires; a flip is a Supposedly Irrelevant Factor that matters. This converts Chen et al.’s footnote into the central, identified test.

(3) **Bin over Z .** Test GARP/SEU *within* each frame and test *cross-frame* preference stability; significant cross-bin instability is the rejection.

(4) **Inference, power, noise.** ARP inference on the RP inequalities; a Bronars random-choice benchmark; statistical RP for stochastic decoding (temperature > 0). This is exactly why a raw CCEI is insufficient.

(5) **Exploitability as the welfare object.** Quantify violations as *money-pump* losses in dollars: construct the sequence of trades a counterparty could offer to extract a sure loss, and measure it. This is the economically meaningful severity, and it speaks to manipulability in markets.

(6) **Welfare.** Apply Bernheim–Rangel: if choices are frame-invariant, welfare analysis on the AI’s decisions is well-defined; if not, characterize the ambiguity. Frame the paper as “*the welfare economics of delegated decisions*,” not “measuring AI rationality.”

Feasibility. *Pilotable now* in your existing revealed-preference notebook: query several models over randomized budgets $\times$ randomized frames, run GARP/RP-bounds, add ARP inference and a money-pump metric. A preliminary figure would be a strong SOP signal.

Verdict (Type A/B). *Type B*: “AI is hyper-rational” is an artifact of sanitized tasks and missing inference; under realistic, adversarial frames with correct inference, agents show systematic, exploitable reversals — so welfare analysis on their choices is unsafe and they are manipulable. *Type A*: violations are pure decoding noise, the systematic component is rationalizable, and rationality *emerges with scale* — a clean licence to treat AI choices as stable preferences.

Intuition — Concretely: the inequalities, the moment, and what Z is here

The inequalities. Give the AI the same budgets a human consumer would face and record its chosen bundles; GARP is the *no-cycle* inequality system (Afriat) — can a single utility rationalize all of its choices? *What Z is.* Z is a *randomized, payoff-irrelevant frame*: the

same budget problem reworded, reordered, or wrapped in different surface context. By construction Z changes nothing real about the decision. *The moment / what is tested.* Test $\mathbb{E}[Z \cdot (\text{change in the chosen bundle})] = 0$: a rational agent's choice is *invariant* to Z , so the bundle should not move with the frame. If it does — systematically, under ARP inference — the AI's “preference” is made by the frame, not stable. *Money pump.* Finally, price the violations: if the choices cycle, there is a sequence of trades (buy, swap, swap, ... back to the start but poorer) that extracts a guaranteed loss. Its size in dollars measures how exploitable — and how far from rational — the agent is.

4.1.4 A worked example: how the test could be run (illustrative)

One concrete design — and it is pilotable today — not the only possible approach.

Data you generate. Query a model on many budget problems: “You have M tokens; good A costs p_A , good B costs p_B ; choose a bundle.” Randomise the prices to trace out budgets (as in any revealed-preference experiment) and record the chosen bundle each time.

The instrument Z (here a multi-category randomized frame). Present each budget under K *payoff-identical* framings $Z \in \{1, \dots, K\}$, assigned at random: reorder the goods; rename them to neutral labels (“good 1”/“good 2”); quote prices as “tokens per unit” vs “units per token”; change number formatting. Using $K > 2$ tests invariance against a whole *family* of irrelevant changes and gives more power.

Step 1 — rationality per frame. Within each frame, check GARP (no choice cycles; Afriat) and compute the CCEI.

Step 2 — the invariance moment. Test $\mathbb{E}[Z \cdot (\text{chosen bundle})] = 0$: does the bundle move across payoff-identical frames? Use Andrews–Roth–Pakes inference (and a Bronars benchmark) to separate genuine frame-sensitivity from decoding noise — run each prompt at temperature > 0 , repeated.

Step 3 — price the violations. For any choice cycle, compute the *money pump*: the guaranteed dollar loss a counterparty could extract.

Reading the result. If choices are GARP-consistent *and* invariant to Z , the model has stable preferences and welfare analysis on its decisions is well-defined (Type A). If the bundle flips with the frame (systematically, beyond noise), or cycles are money-pumpable, the “rationality” is a frame artifact and the agent is manipulable (Type B) — so welfare analysis on AI-delegated decisions is unsafe.

4.2 Thread 2 — Does market competition still discipline discrimination once decisions are algorithmic?

4.2.1 The theory

Becker's deductive prediction, and the new twist

Becker (1957) proved that *taste-based* discrimination is paid out of *rents*: under competition, non-discriminating firms out-compete discriminators, so product-market competition should *erode* taste-based discrimination — but it should *not* erode *statistical* discrimination (using group averages to predict, in the tradition of Arrow and Phelps), which requires no taste and no rents. The new question: *algorithms* can sustain statistical discrimination cheaply and without rents — so does competition still erode discrimination when decisions are *automated*?

4.2.2 Existing work, and the gap

The prediction holds in *human* markets: Hirata–Soares (2016) use Brazil’s trade liberalization (a Kovak/Bartik shift-share) and find a 9.7pp tariff cut $\rightarrow$ an 18% smaller racial wage gap, stronger in concentrated/high-rent sectors; Levine, Levkov & Rubinstein (2008) find the same via U.S. bank deregulation. The algorithmic-discrimination literature, by contrast, *documents and proposes to regulate* bias (Fuster et al. 2022 on machine-learning credit; Bartlett et al. 2022 on FinTech lending; Rambachan–Kleinberg–Ludwig–Mullainathan 2020 on regulating algorithms; Obermeyer et al. 2019), and Calvano et al. (2020) show pricing algorithms can learn to *collude*. *But no one tests Becker’s competition-erosion prediction in algorithmic vs. human markets.*

4.2.3 Your new test (Thread 2)

How to go beyond — Separating taste-based from statistical discrimination via competition $\times$ automation

Interact a *competition shock* (trade exposure, platform entry, deregulation — a shift-share / ACD-style design, the Z) with the *degree of algorithmic decision-making* in a sector, and test whether Becker’s erosion *breaks* where decisions are automated. Because competition erodes taste-based but not statistical discrimination, the interaction *simultaneously* (i) *separates the two mechanisms* — a long-standing identification problem — and (ii) asks whether markets still self-correct discrimination in the AI era. *Type B*: erosion holds in human but not algorithmic markets $\Rightarrow$ markets no longer self-correct $\Rightarrow$ algorithmic audits are justified. *Type A*: erosion survives automation $\Rightarrow$ competition still disciplines it. *Main challenge to defend*: measuring “degree of algorithmic decision-making” by sector.

Why it matters — Domain of Chapter 4

The defining economic transformation: whether AI-mediated markets can be *evaluated* at all (welfare, antitrust, consumer protection), whether AI agents are *manipulable*, and whether delegating hiring and lending to algorithms *breaks the market’s self-correction of discrimination* — the algorithmic-fairness and audit agenda (the EU AI Act; U.S. algorithmic-accountability proposals).

Essential readings — Chapter 4 — rationality of machines

Thread 1: Chen, Liu, Shan & Zhong (2023) — the result you upgrade; Raman et al. (2024, STEER) — the benchmark you differentiate from; Bernheim & Rangel (2009) — the welfare/delegation backbone (re-read with AI in mind); Horton (2023) and Mazeika et al. (2025) — the AI-agent frontier. **Thread 2:** Becker (1957) — taste-based discrimination; Hirata & Soares (2016) and Levine, Levkov & Rubinstein (2008) — competition erodes it in human markets; Fuster et al. (2022) and Bartlett et al. (2022) — algorithmic discrimination documented; Rambachan, Kleinberg, Ludwig & Mullainathan (2020) — the economics of regulating algorithms; Calvano et al. (2020) — algorithmic collusion. *Re-scan this chapter’s literature right before submitting — it moves monthly.*

Chapter 5

Synthesis: One Method, Three Frontiers

5.1 The single contribution behind all three

Across time, risk, and machines you are doing *one* new thing: **building a more robust test of a deductive theory by filtering confounding shocks with an exogenous instrument Z , then doing correct inequality inference**, exactly as Adão–Costinot–Donaldson do for trade models and as you do with GARP conditional on Z . The three theories are applications; the method — nonparametric revealed preference made decisive by clean identification, with a Bernheim–Rangel welfare backbone — is the through-line. The table summarizes the common skeleton.

	Time	Risk	Machines
Theory	Becker–Murphy rational addiction	Expected utility (+ Rabin)	Utility max / SEU by AI agents
Behavioral rival	β - δ present bias; temptation	Prospect theory / prob. weighting	Frame-dependence; statistical discrimination
RP test	Crawford; Blow– Browning–Crawford	Polisson–Quah– Renou GRID; Echenique–Saito	GARP/Afriat; Echenique–Saito SEU
Nuisance	serially-correlated tastes	selection into gam- bling/insurance	prompt/frame $\sim$ task
Instrument Z	announced taxes; legalization	betting legalization	randomized payoff-irrelevant frame; competition shock
Inference	ARP + Bronars	ARP + efficiency index	ARP + money-pump
Welfare hook	sin tax vs. design (internalities)	insurance/justice	is welfare analysis on AI defined?

5.2 How to go beyond, in one paragraph (reusable)

For any of the three: take the theory’s revealed-preference inequalities; find an exogenous Z that moves the relevant price/availability but is orthogonal to the nuisance; test the inequalities *conditional on Z* by binning; use ARP for correct size and Bronars for power; and read the verdict through Bernheim–Rangel for welfare. The novelty is never “a new RP test” in the abstract (those exist); it is *the identification* — bringing a credible, exogenous Z to a nonparametric rationality test in a high-stakes field (or AI) setting where it has policy bite.

5.3 Prioritization

1. **Risk (Ch. 3)** — the *cleanest* novelty: the method (GRID) exists and explicitly invites the field application; the instrument (betting legalization) is clean; the insure-and-gamble field test is genuinely untested.
2. **Time (Ch. 2)** — the *closest analog to your GARP-conditional-on- Z* work, with the largest policy resonance; pitch as identification + policy, since Blow–Browning–Crawford and Allcott et al. occupy nearby ground.
3. **Machines (Ch. 4)** — the *most original and most current*, and *pilotable now*; Thread 1 must be framed as “welfare of delegated decisions” (not “measuring AI rationality,” which is crowded); Thread 2 (competition $\times$ automation) is cleaner and less occupied.

5.4 Cross-links worth stating in the SOP

The three are not three topics but one program: (i) the *same* revealed-preference toolkit tests all of them; (ii) “option luck” (deliberate gambling) in the risk chapter connects to the distributive-justice and equality-of-opportunity themes; (iii) AI is not a fourth theory but a new *domain* in which the time/risk/utility axioms can all be re-tested — and where the welfare stakes (can we evaluate delegated decisions?) are highest. Lead the SOP with the method identity; centre one study (risk or machines); be candid about its nearest competitor and your one-sentence differentiator; keep a policy hook in every paragraph.

Appendix A

Reading Guide, Glossary, and References

A.1 A consolidated reading order (if you read nothing else)

This is the curated path through the primary sources — the “option A” list, in order. (I can stack these into a single PDF for your e-reader on request.)

Essential readings — Tier 1 — the eight to read first

(1) Echenique (2020) — the RP ladder (method, short). **(2)** Adão–Costinot–Donaldson (2025) §I–II — the two-forces logic and the Z test. **(3)** Bernheim & Rangel (2009) §I–II — welfare without rationality. **(4)** Becker & Murphy (1988) + Gruber & Köszegi (2001) — the time theory and its identification. **(5)** Rabin (2000) + Kahneman & Tversky (1979) — the risk theory and its rival. **(6)** Polisson–Quah–Renou (2020) — the risk RP method (your tool). **(7)** Blow–Browning–Crawford (2021) — the time RP method (your nearest competitor). **(8)** Chen et al. (2023) + Raman et al. (2024) — the AI results you build on and differentiate from.

Essential readings — Tier 2 — depth and foils

Chambers & Echenique (2016, book); Andrews–Roth–Pakes (2023); Auld & Grootendorst (2004); Dragone & Raggi (2020); Crawford (2010); Allcott–Gentzkow–Song (2022); Allcott–Lockwood–Taubinsky (2019); Barseghyan et al. (2013, and 2018 survey); Echenique–Imai–Saito (2019); Choi–Kariv–Müller–Silverman (2014); Barberis (2013); Köszegi–Rabin (2006); Becker (1957); Hirata–Soares (2016); Levine et al. (2008); Fuster et al. (2022); Bartlett et al. (2022); Rambachan et al. (2020); Calvano et al. (2020); Baker et al. (2024).

A.2 Glossary

Deductive theory

A prediction derived from assumptions (a theorem), refutable by data — as opposed to an estimated regularity.

GARP

Generalized Axiom of Revealed Preference: choices are rationalizable by stable utility iff they contain no preference cycle (Afriat’s theorem).

Afriat inequalities

The finite linear system whose feasibility is equivalent to rationalizability; the template

for all stronger tests.

CCEI

Critical Cost Efficiency Index: budgetary “waste” needed to erase GARP violations; 1 = perfect. No sampling theory by itself.

RP bounds

Nonparametric prediction set for a new choice, using rationality alone.

Nuisance shocks

All forces other than the one the theory is about; they contaminate naïve tests.

Instrument Z

Variation correlated with the force of interest but orthogonal to the nuisance; isolates the former.

Shift-share IV

An instrument from exogenous “shifters” $\times$ baseline “shares” (ACD; Bartik; Kovak).

Binning

Partitioning observations by Z to test a conditional restriction as finitely many unconditional ones.

Moment inequalities / ARP

Theory-implied “ $\geq$ ” restrictions; tested with Andrews–Roth–Pakes (2023) inference, not regression.

Bonferroni power

Benchmark for a revealed-preference test: how often random choices would fail it.

Statistical/stochastic RP

Revealed-preference testing that accounts for noise/measurement error (Chambers–Echenique; Varian 1985).

Adjacent complementarity

Becker–Murphy prediction: anticipated future prices affect current consumption.

Present bias (β - δ)

Over-weighting the present; time inconsistency; demand for commitment.

Expected utility (EU/SEU)

Probability- (or belief-) weighted average utility; risk aversion from concavity.

Probability weighting

Prospect-theory distortion of probabilities (overweighting small ones); resolves Friedman–Savage.

Rabin’s theorem

Small-stakes risk aversion logically forces absurd large-stakes aversion under $EU \Rightarrow EU$ refuted.

GRID

Polisson–Quah–Renou’s nonparametric test of EU and its generalizations; admits risk-loving.

Ancillary condition / SIF

A payoff-irrelevant feature (framing, timing, salience) that may move choice; Bernheim–Rangel’s “ancillary condition” = Thaler’s “supposedly irrelevant factor.”

Money pump

A sequence of trades extracting a sure loss from an inconsistent agent; dollar measure of violation severity.

Taste-based vs. statistical discrimination

Discrimination from animus (eroded by competition) vs. from group-average prediction (not eroded).

Type A / Type B

Rehabilitating a theory naïve tests rejected / debunking a theory naïve tests confirmed.

Companion materials

The full analysis pipeline (Monte-Carlo, experimental and field arms, figures) lives in `extt-twritng_sample/TIME_RATIO/code/final/` with a README that maps every number in this book to the script and output file that produced it. Two planning notes accompany this chapter: `extttSOP_ideas/new_data_assessment_2026_07_10.md` (the audit of every dataset on disk) and `extttSOP_ideas/phase_B_next_steps.md` — the vetted plan for the next data extension (a cue-rich session-level panel from the Lichess open database, the Goodreads Book Graph want/should design, and the ranked alternatives), kept out of the main text because it is a roadmap, not a result.

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